

Discrimination Metrics

Matt Engelhard

Preliminaries

Metrics should be informed by the clinical use case

Medical student case study: new kidney risk prediction model

You are a primary care provider seeing diabetic patients. You learn that some colleagues have implemented a new prediction model that aims to prevent or delay kidney failure by predicting 2-year risk of progression to end-stage chronic kidney disease.

This model is based on electronic health record data, including diagnosis and procedure codes, medications, labs, and vitals; as well as social determinants of health.

You think the model sounds promising and start thinking about ways you might use it in practice.

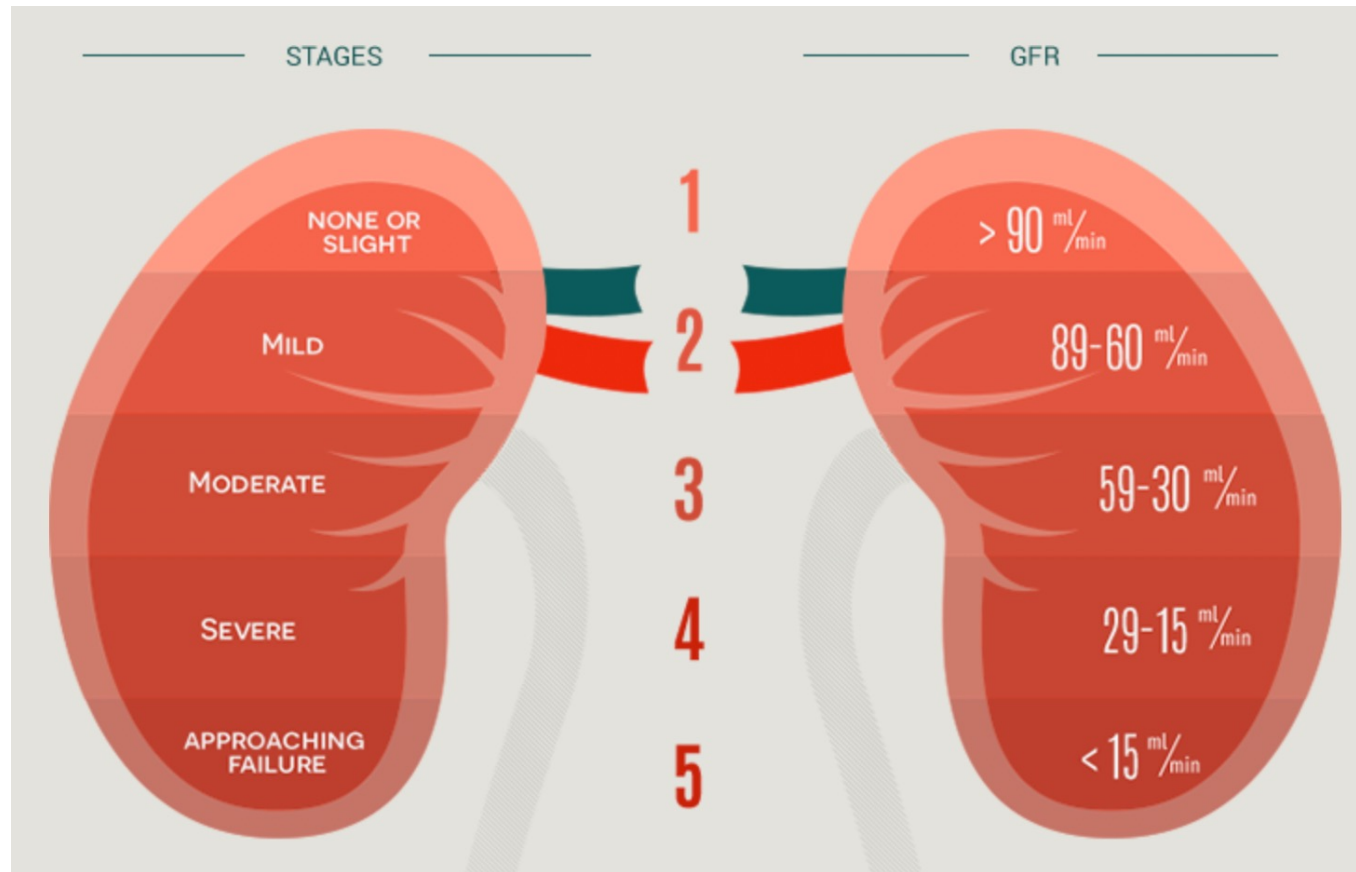
Health AI student case study: new kidney risk prediction model

You are a health AI student developing a new prediction model that aims to prevent or delay kidney failure by predicting 2-year risk of progression to end-stage chronic kidney disease.

This model is based on electronic health record data, including diagnosis and procedure codes, medications, labs, and vitals; as well as social determinants of health.

You want to design a performance evaluation that is aligned with the way providers will use your model in practice.

Consider: how might a provider use your model's predictions?



11%

Probability of Stage IV CKD within two years

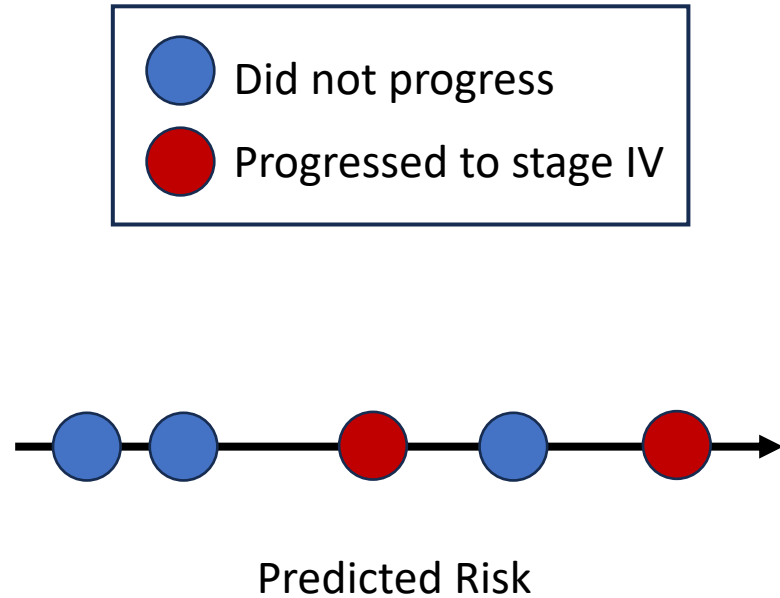
Report Back:

- How might a clinician use this number (be specific)?
- What performance metrics are relevant to this way of using the prediction?

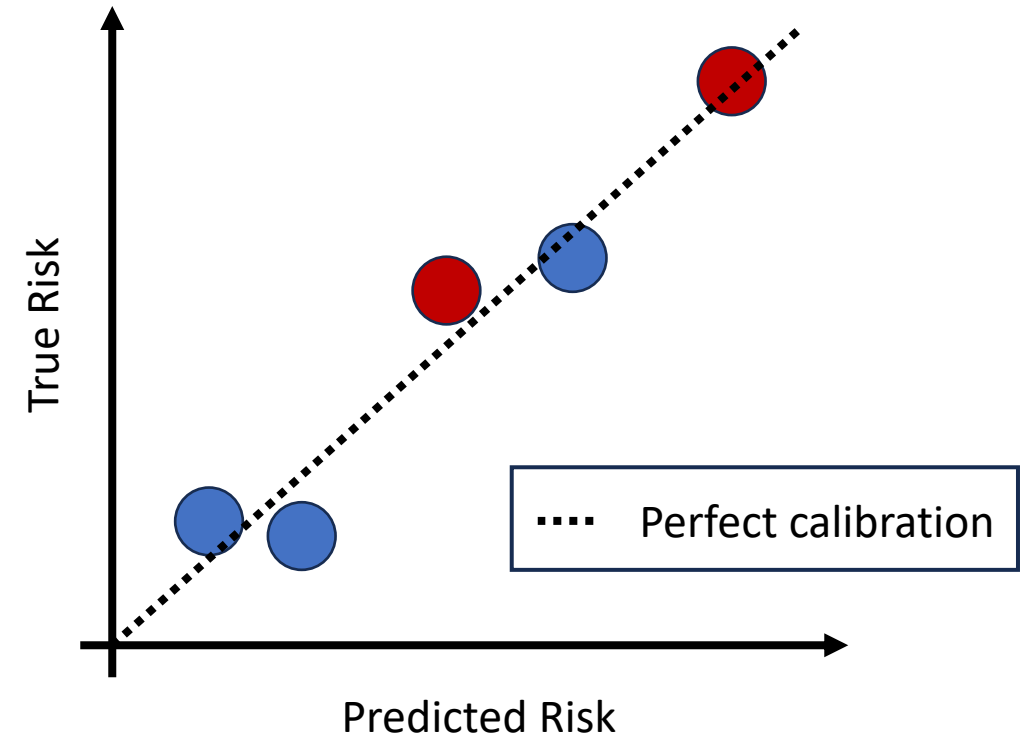
Discrimination

vs

Calibration



Can we identify individuals who will progress?



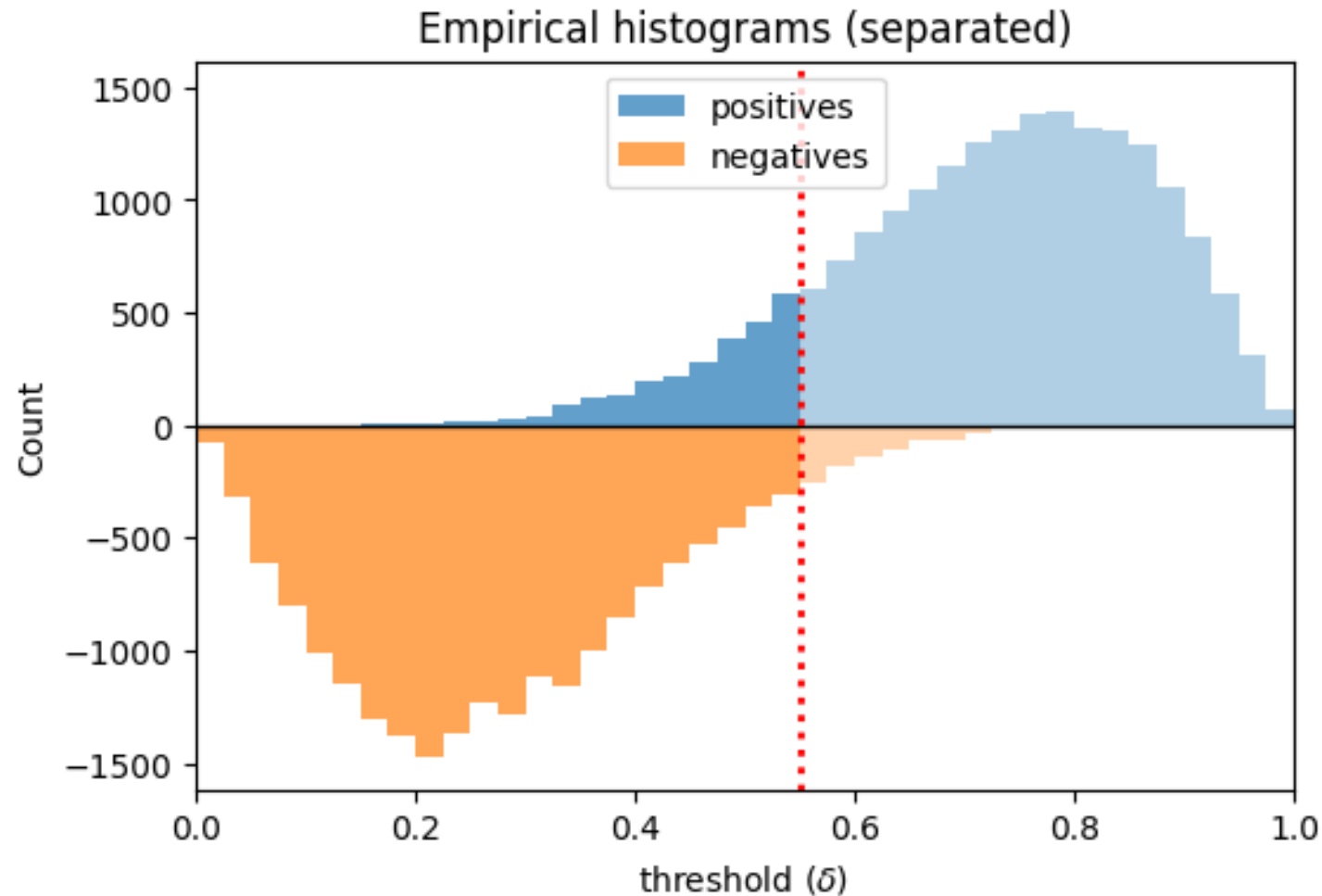
Are our predicted probabilities accurate?

Today: discrimination metrics

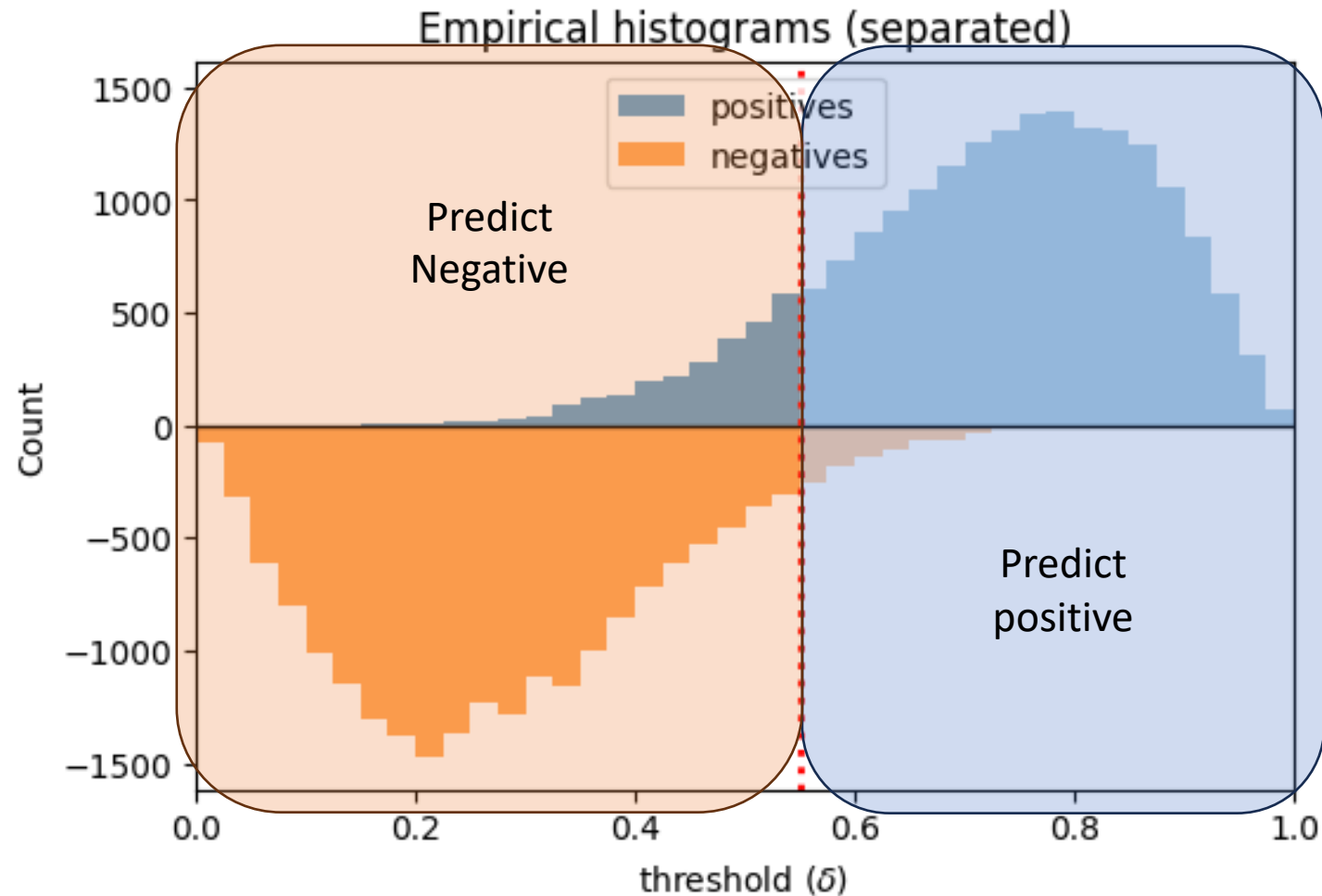
Next time: calibration and class balance

Suppose our features are highly informative.
What might our model's predictions look like?

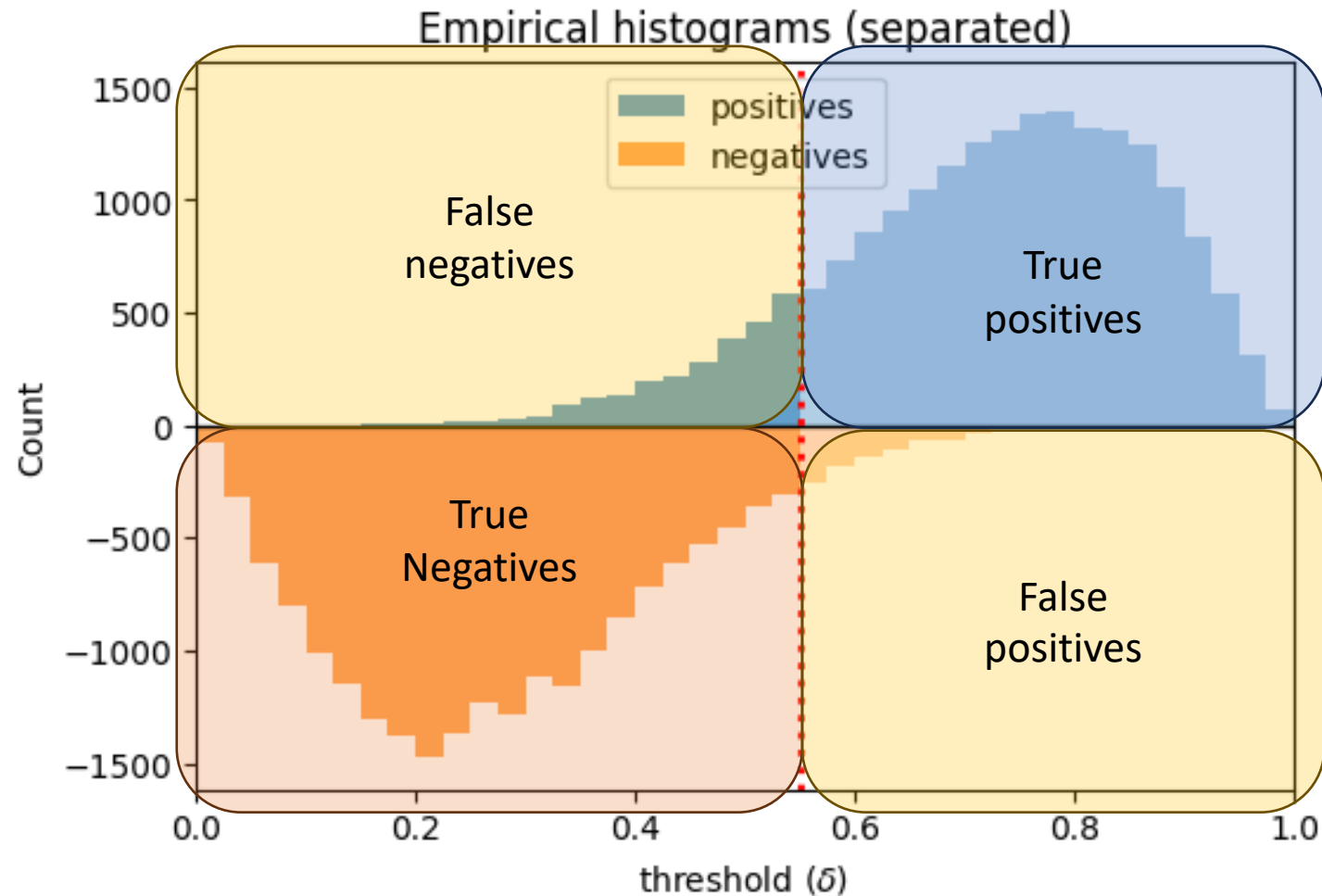
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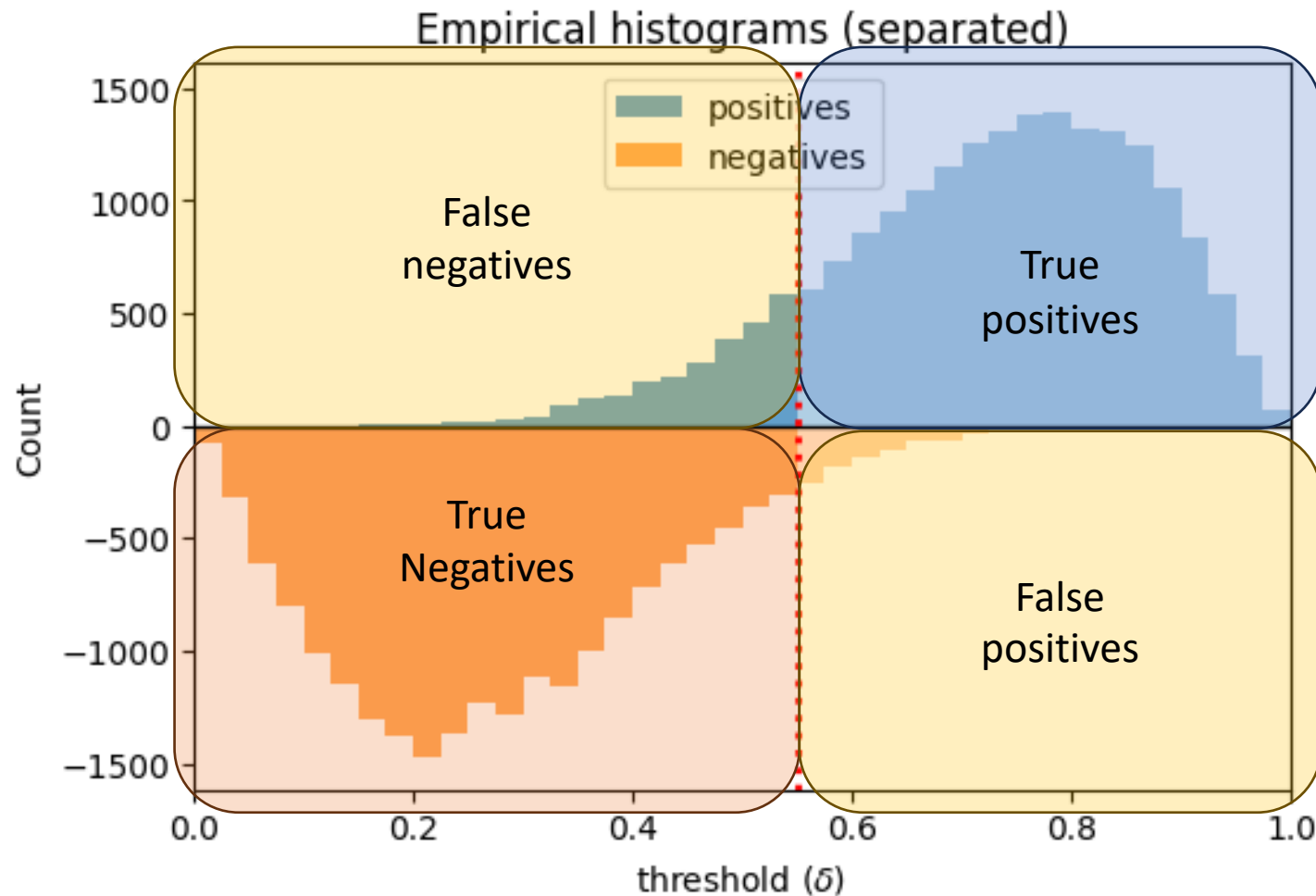
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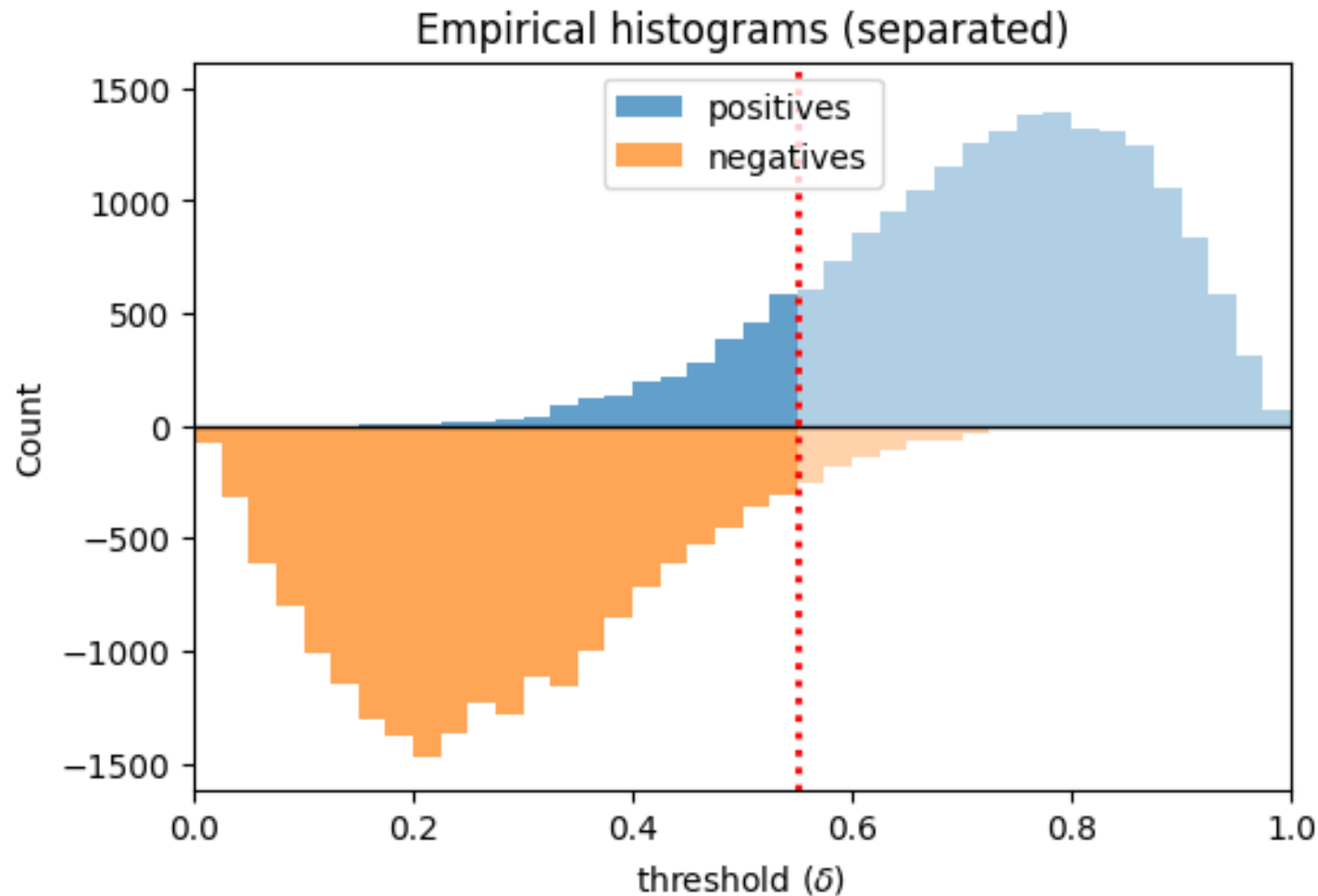


Review threshold-specific metrics:

- TPR (sensitivity, recall)
- FPR (1 – specificity)
- Precision (PPV)
- NPV

Consider the clinical utility of each!!

Consider: how does each metric change as we vary the threshold?

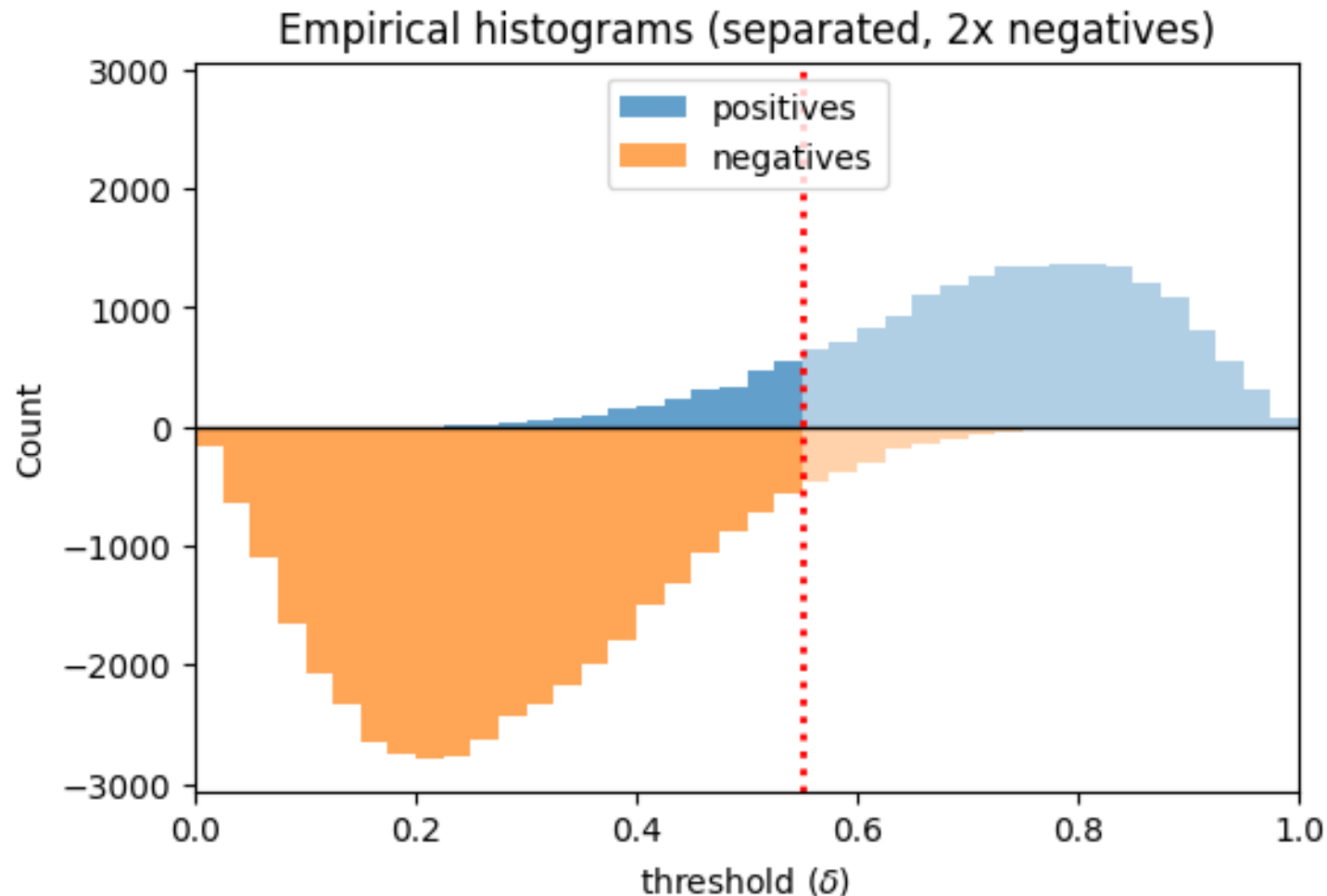


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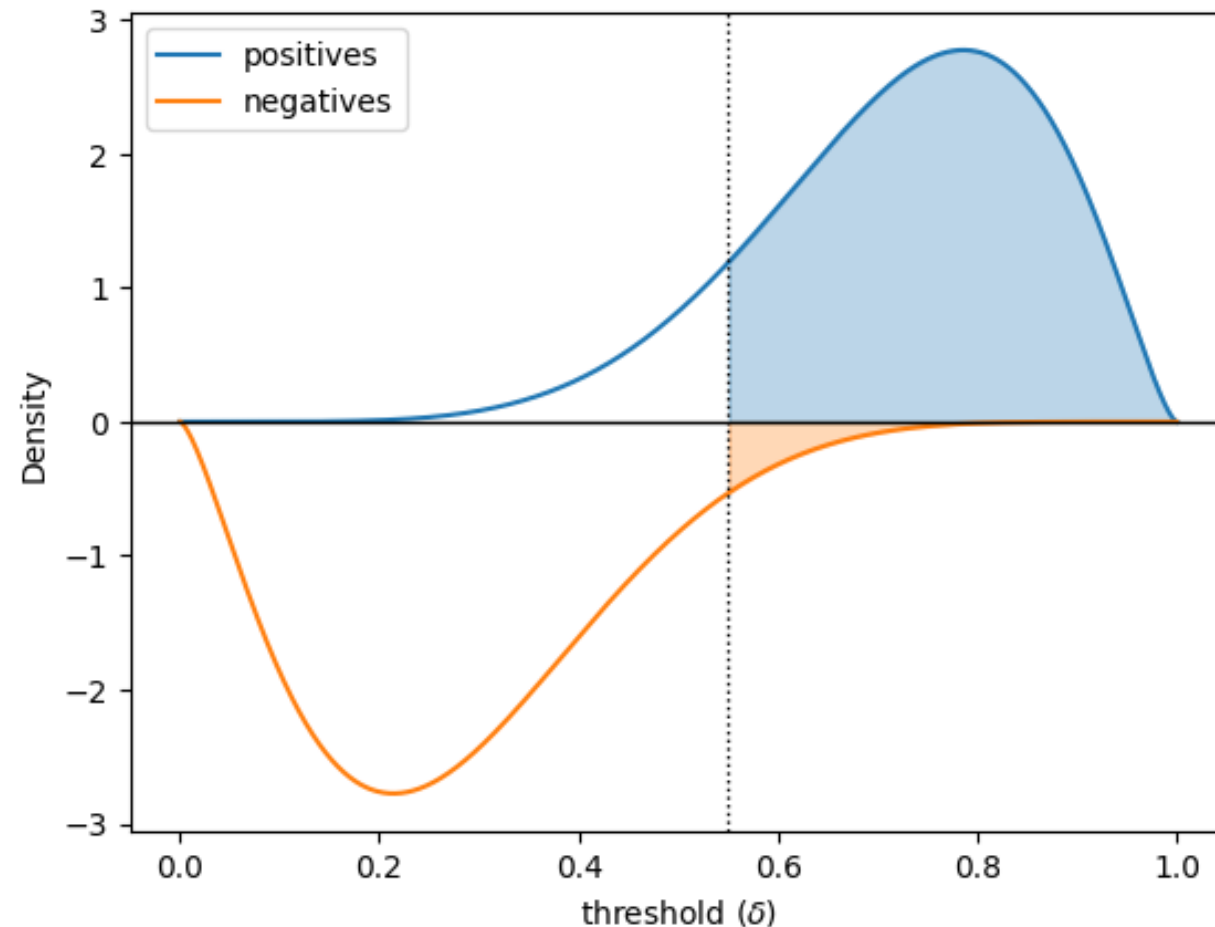


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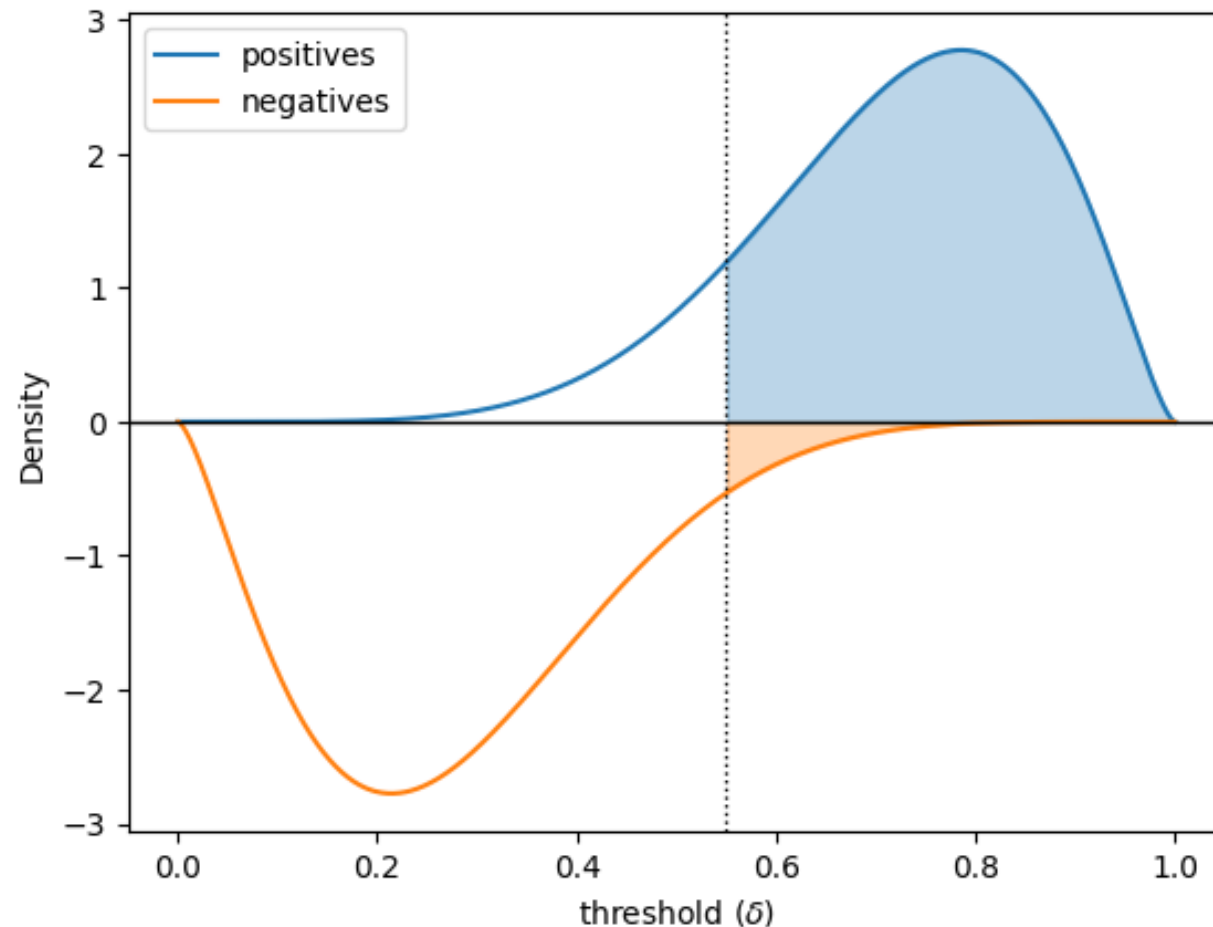
Let S_+ and S_- denote random variables associated with model predictions for positive and negative outcomes, respectively.

Note that:

$$\text{TPR}(\delta) = P(S_+ > \delta) = 1 - \underbrace{P(S_+ \leq \delta)}_{\text{CDF for positives}}$$

$$\text{FPR}(\delta) = P(S_- > \delta) = 1 - \underbrace{P(S_- \leq \delta)}_{\text{CDF for negatives}}$$

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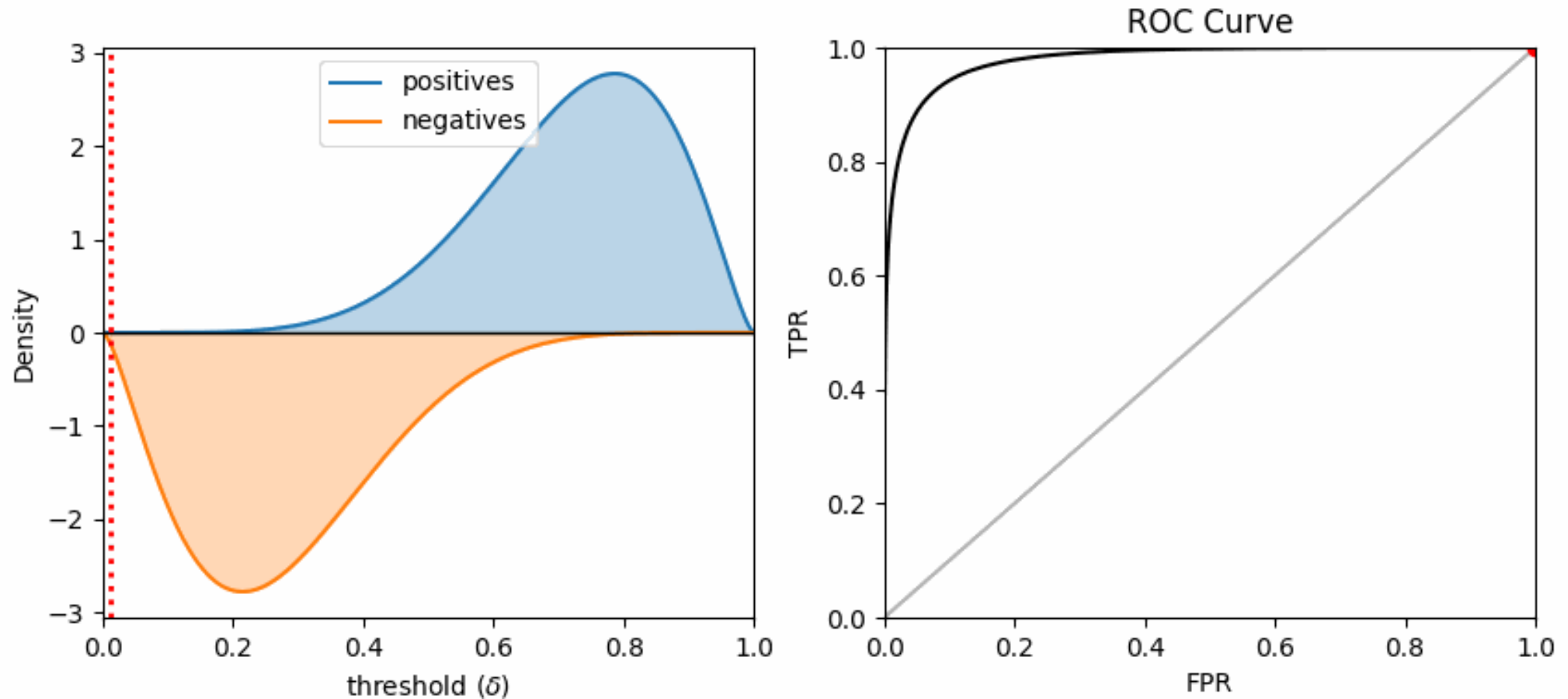
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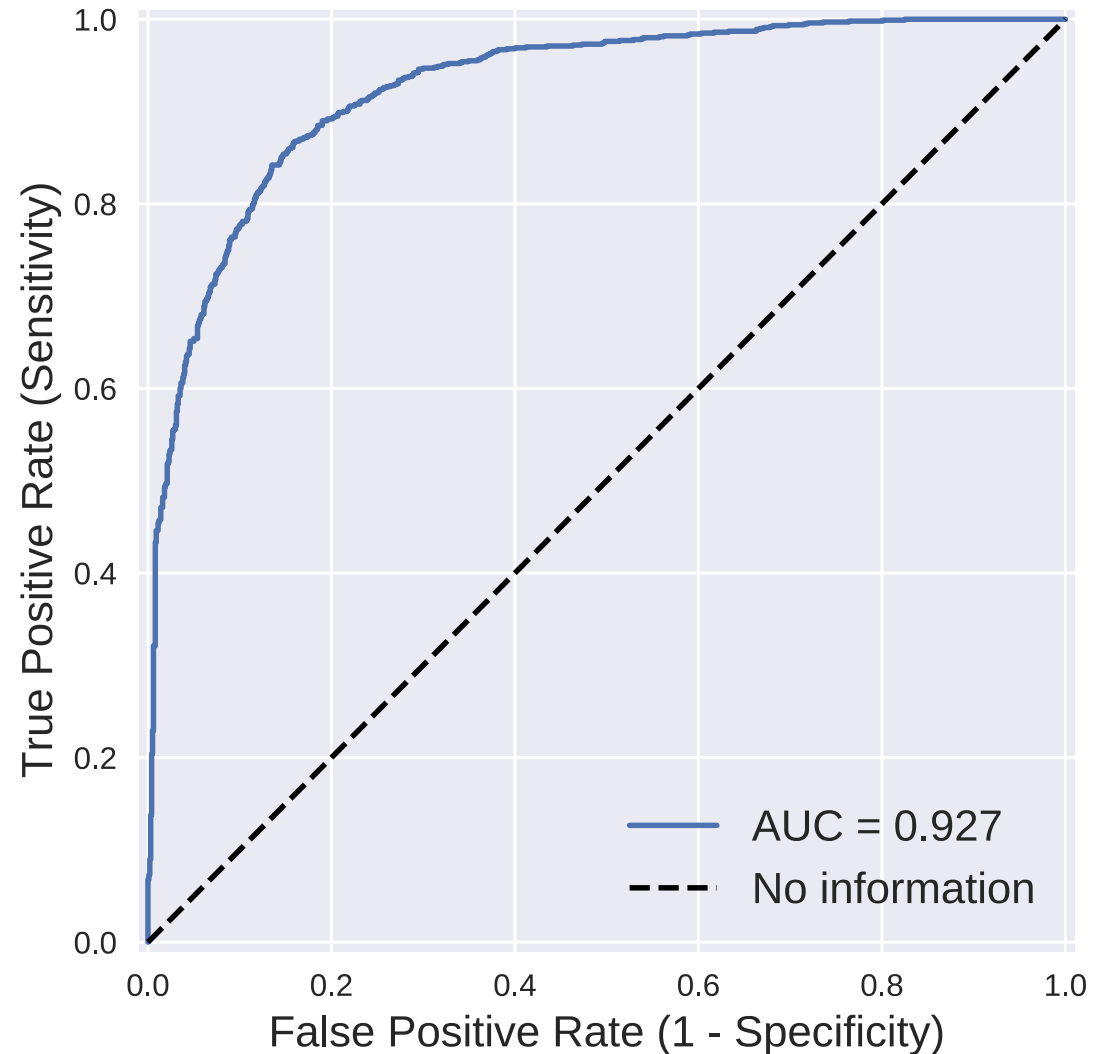
TPR & FPR decrease monotonically as δ increases

Consider: how does each metric change as we vary the threshold?



Receiver Operating Characteristic Curve

- Illustrates the tradeoff between the TPR (sensitivity) and FPR (1–specificity) as we vary the threshold.
- Allows clinicians to consider this tradeoff and select appropriate operating points
- The area under this curve (AUC) provides a single number summarizing this tradeoff.
- In most cases this is the most important metric (why??)

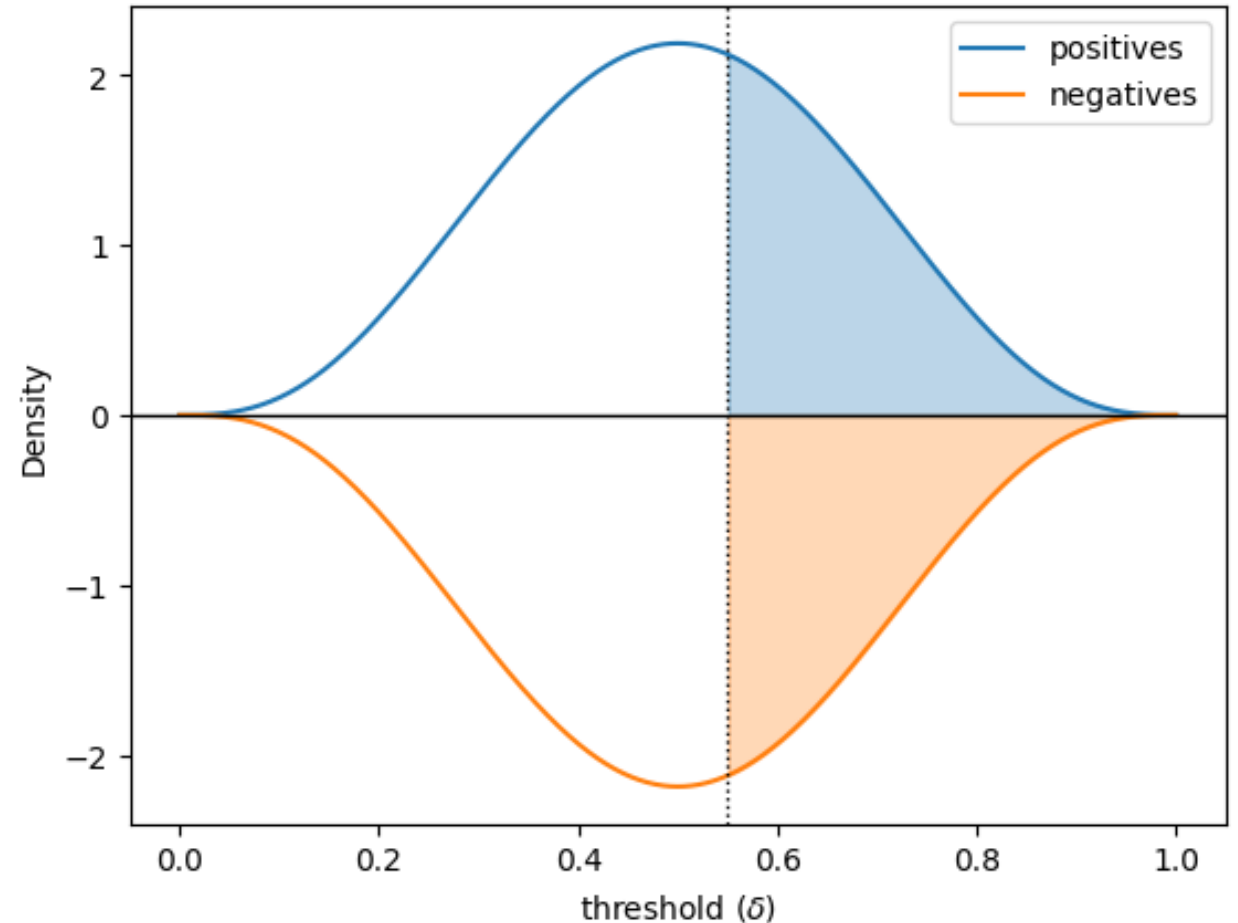


So, what's a *good* AUC value?
(i.e., *good* performance)?

We'll start to answer this question by taking a look at *bad* performance.

Suppose our features contain no information about the outcome. What might predictions look like?

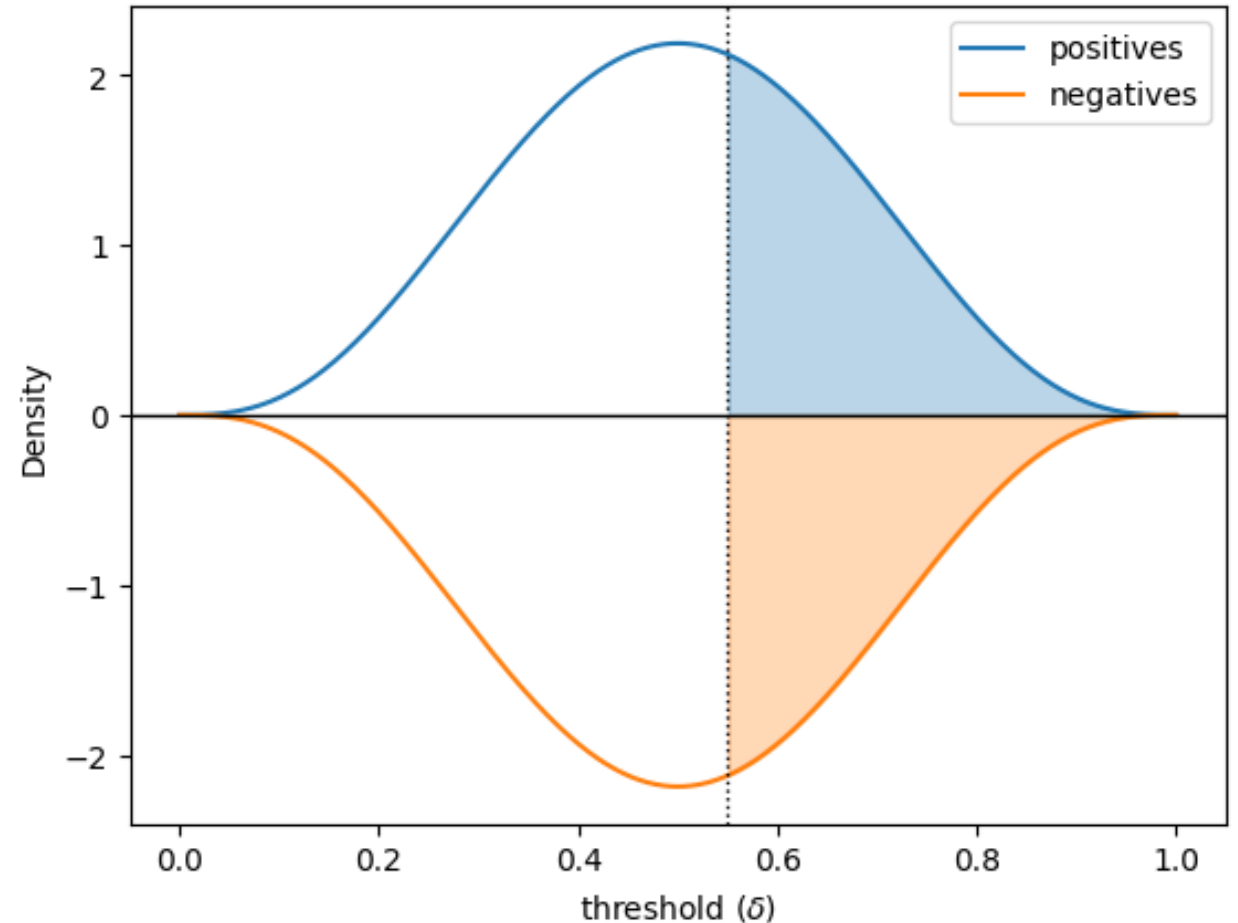
- Similar distributions between positive and negative cases.
- The predicted value tells you nothing about which one it's more likely to be.



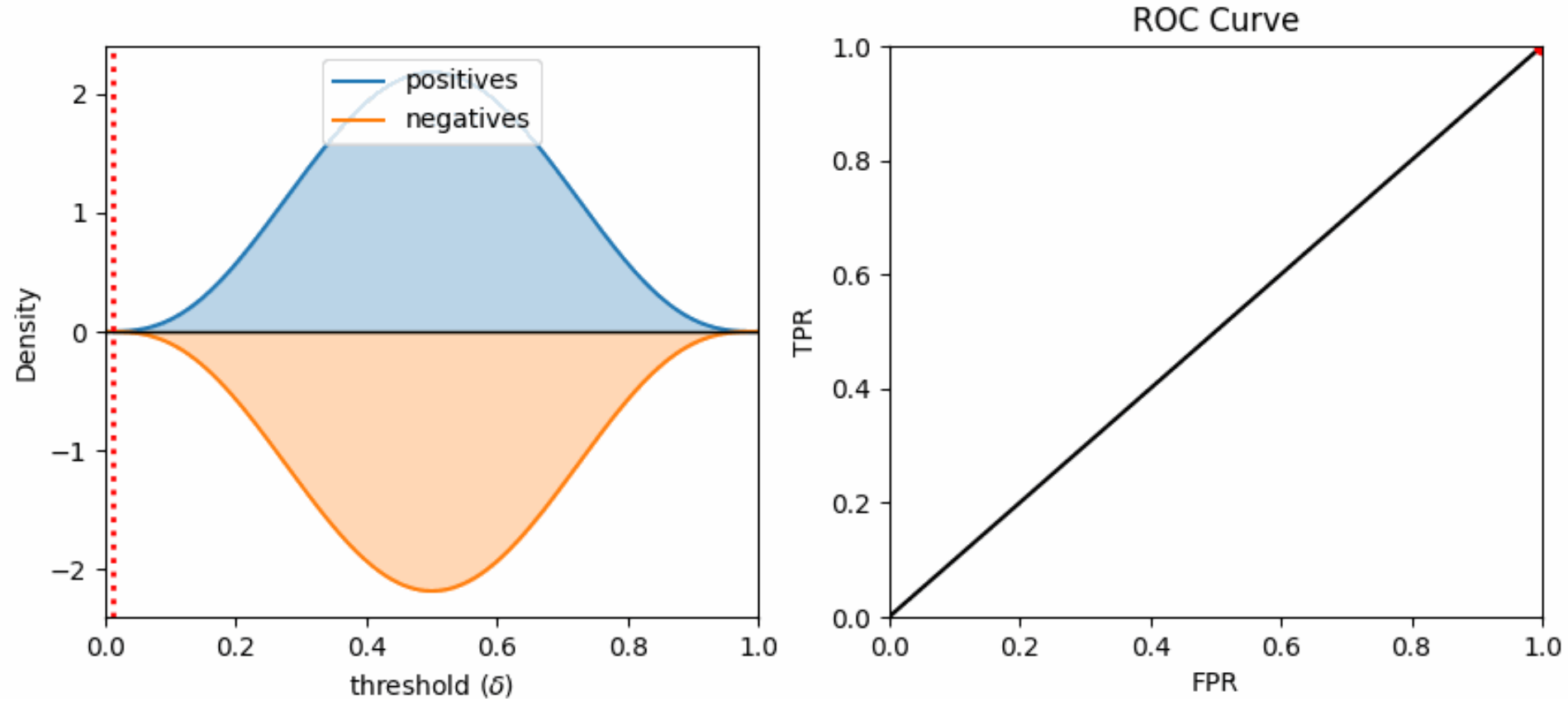
Suppose our features contain no information about the outcome. What might predictions look like?

What is the:

- (a) TPR (sensitivity)?
- (b) FPR ($1 - \text{specificity}$)?
- (c) PPV?



Our “no information” predictive model

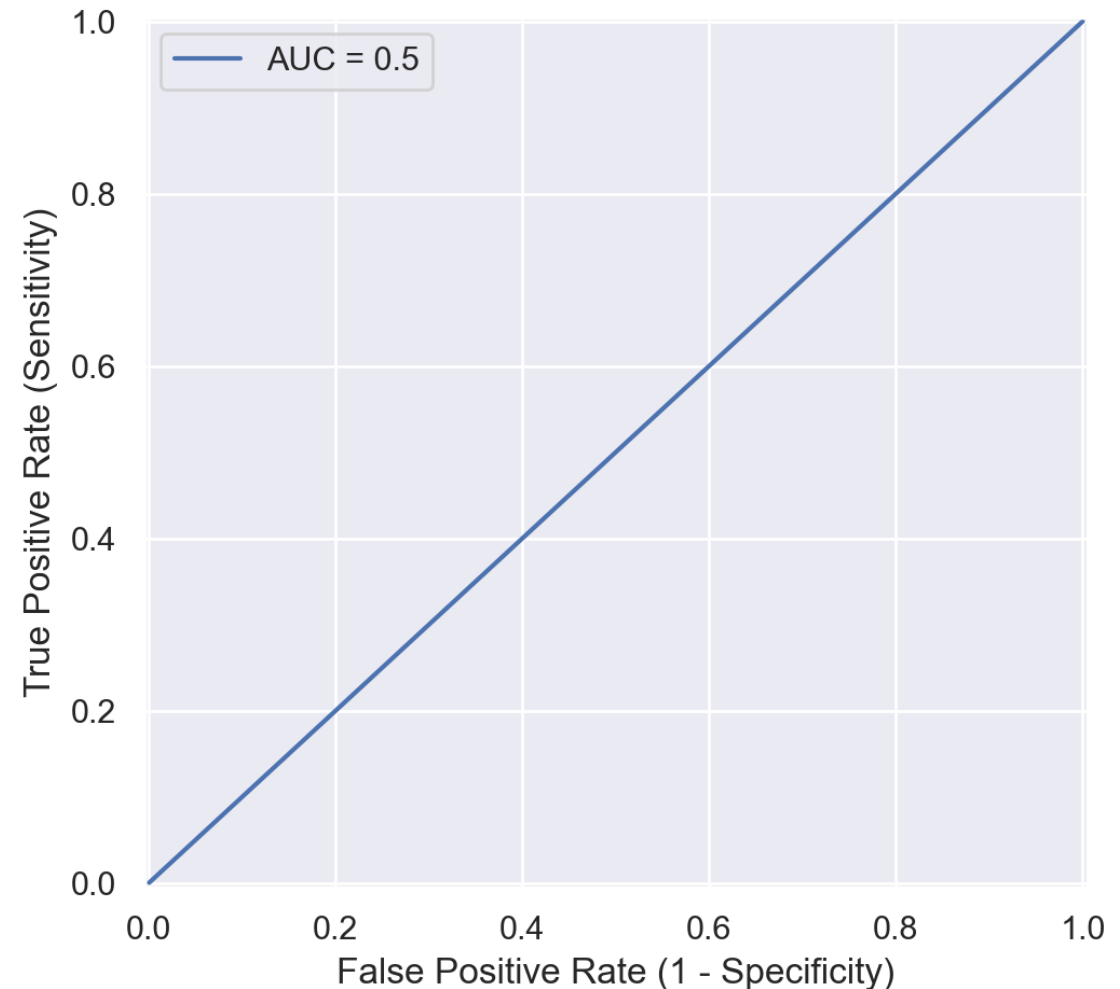


TPR and FPR always match (why?)

- the distributions of positive and negative cases match
- the corresponding CDFs match

Our “no information” model

- AUC is 0.5
- Do we ever see $AUC < 0.5$?
- If so, what can we infer about the distributions of our predictions for positive and negative samples?



So, what's a *good* AUC value? (i.e., *good* performance)?

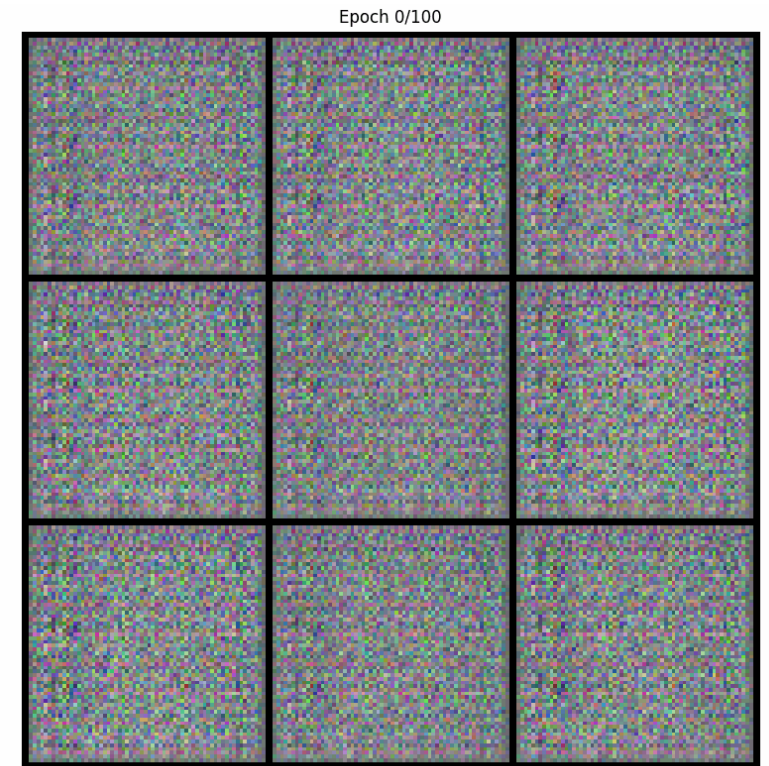
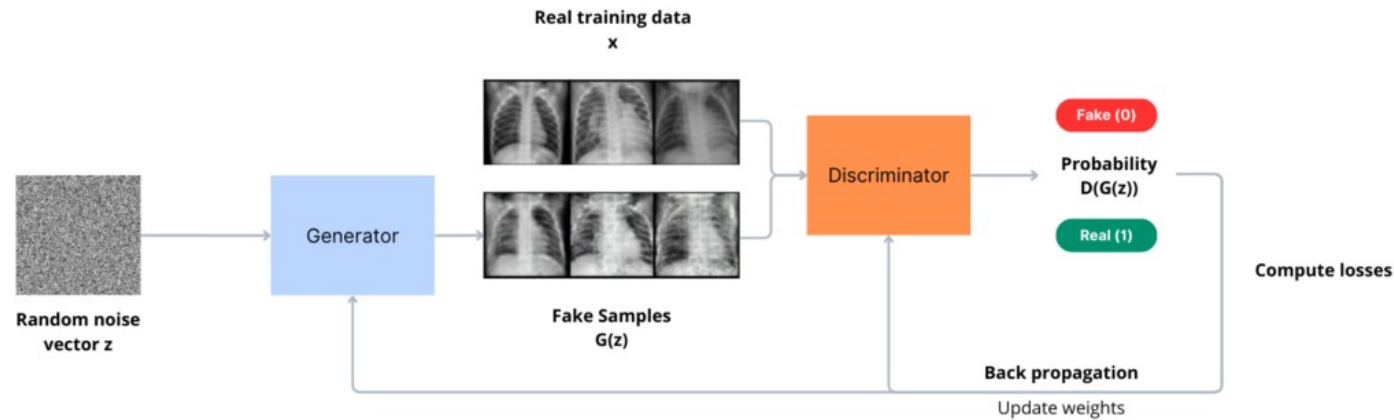
It depends.

- Are predictions better than random?
- Are predictions better than the previous best performing model?
- Are predictions better than expert performance?
- Does performance exceed our (informed) expectations?
- **Is the model clinically useful?**
- Different expectations for risk prediction vs classification

What can we say about prediction performance when distributions match?

- When distributions of predictions match between pos/neg:
 - predictions provide no information about the outcome
 - i.e. random guessing
- When distributions of features (observed or latent) match between pos/neg:
 - features contain no information about the outcome
- Note that adversarial methods operate on this principle:
 - Train a *discriminator* that predicts pos/neg (or real/fake)
 - Train a *generator* whose goal is to increase the discriminator's loss (many variations on this)
 - Result? Distribution matching

Brief overview of generative adversarial networks (GANs)



$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))]$$

Adversarial discriminative domain adaptation (ADDA)

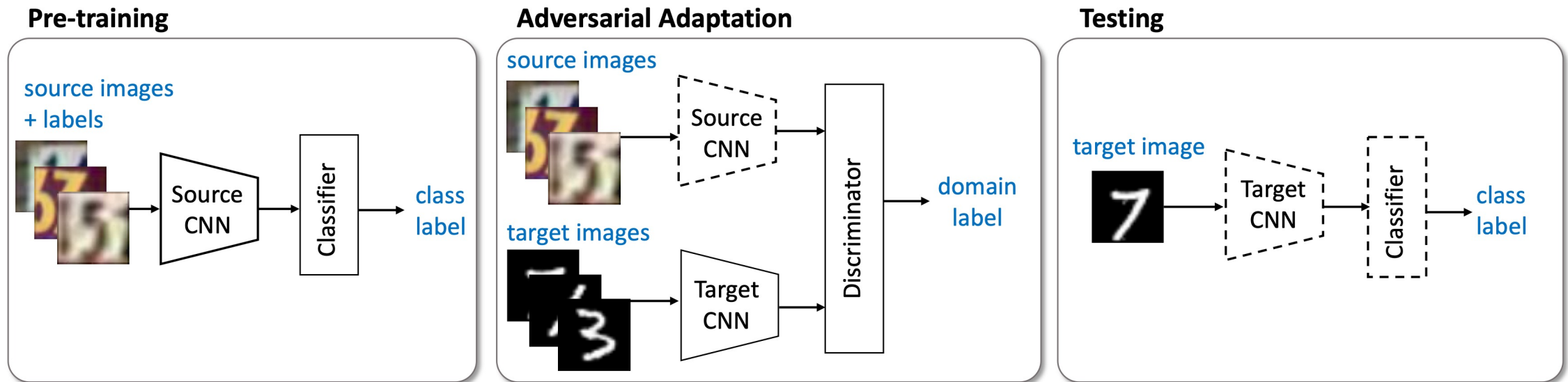


Figure 3: An overview of our proposed Adversarial Discriminative Domain Adaptation (ADDA) approach. We first pre-train a source encoder CNN using labeled source image examples. Next, we perform adversarial adaptation by learning a target encoder CNN such that a discriminator that sees encoded source and target examples cannot reliably predict their domain label. During testing, target images are mapped with the target encoder to the shared feature space and classified by the source classifier. Dashed lines indicate fixed network parameters.

We've quantified performance across all thresholds, but how do we use the model?

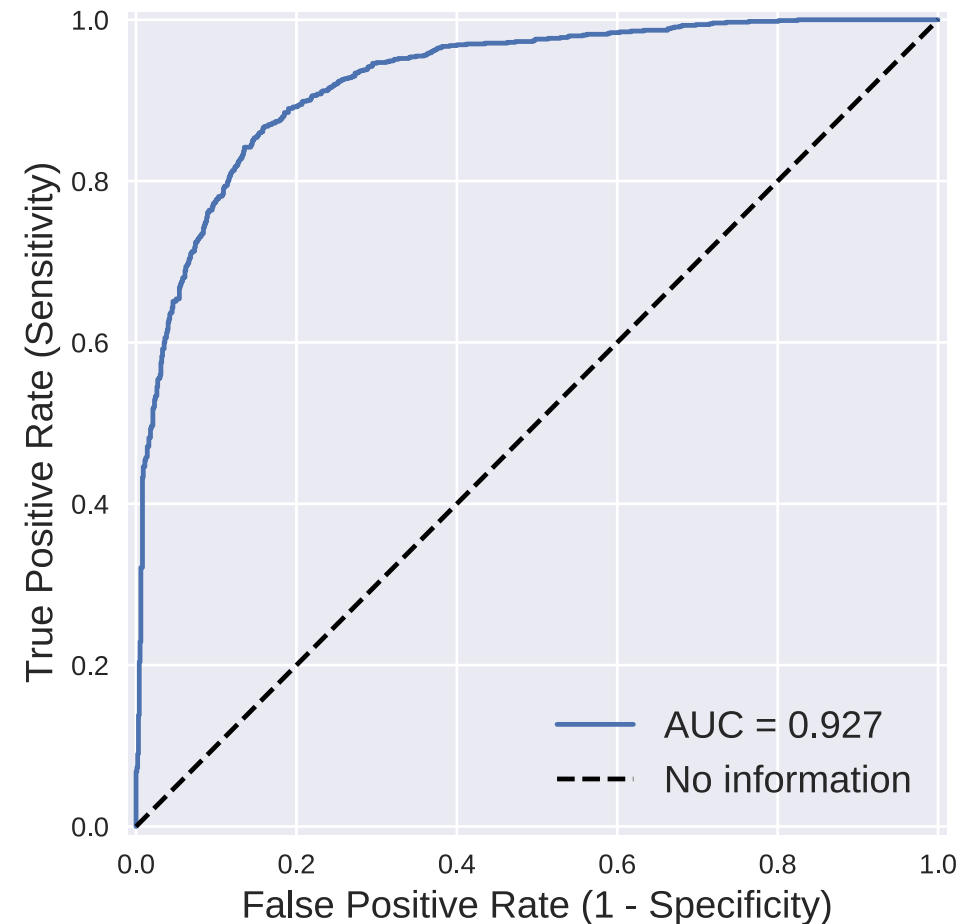
Sometimes the predicted probability really is what we care about.

- *Example*: probability of heart attack
- If so, we need to make sure our model is *calibrated*

More often, we need to pick a threshold so we can decide whether to:

- Alert a provider
- Get a biopsy
- Refer the patient
- etc

What threshold should we pick? What's the right tradeoff?

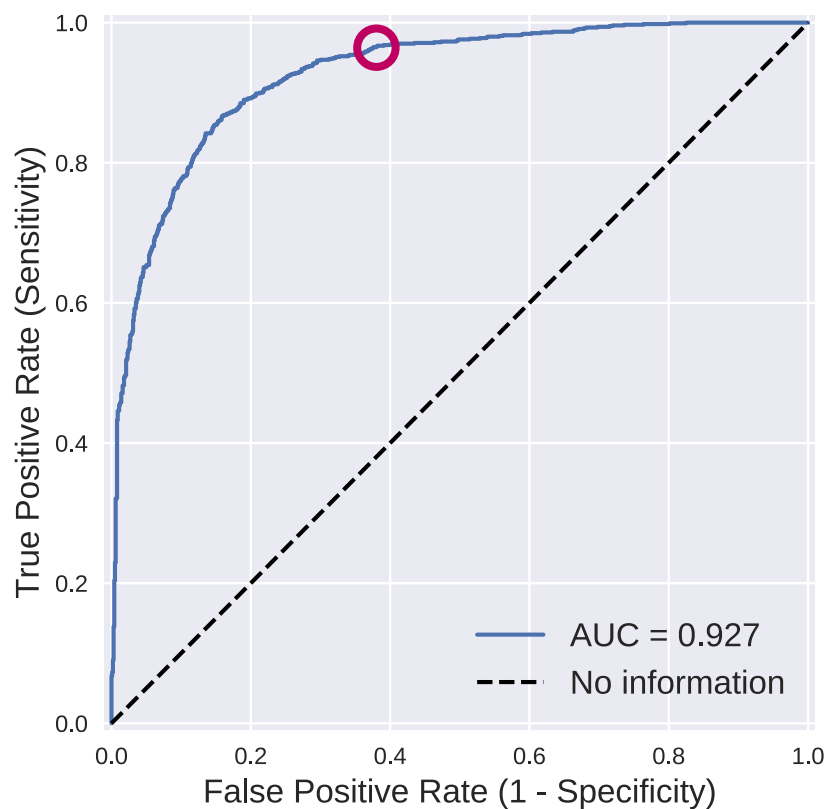
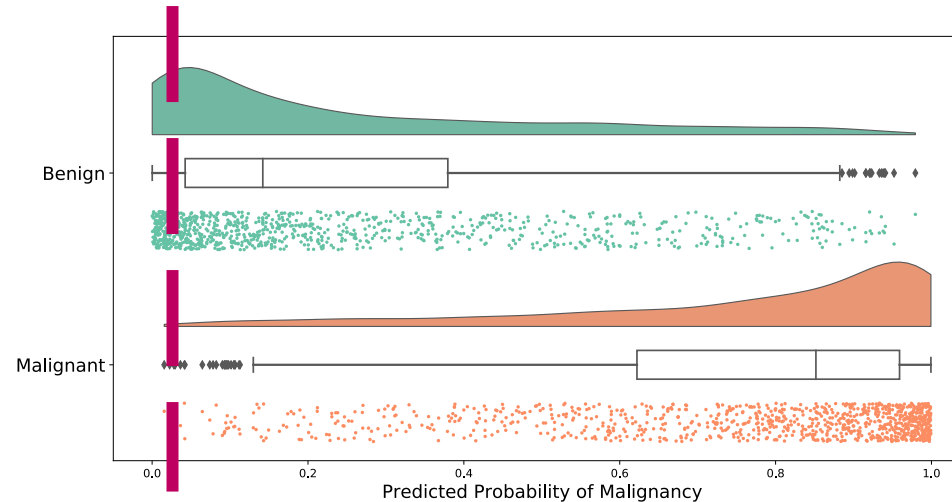


Healthcare Scenarios

Which performance metric is most important?

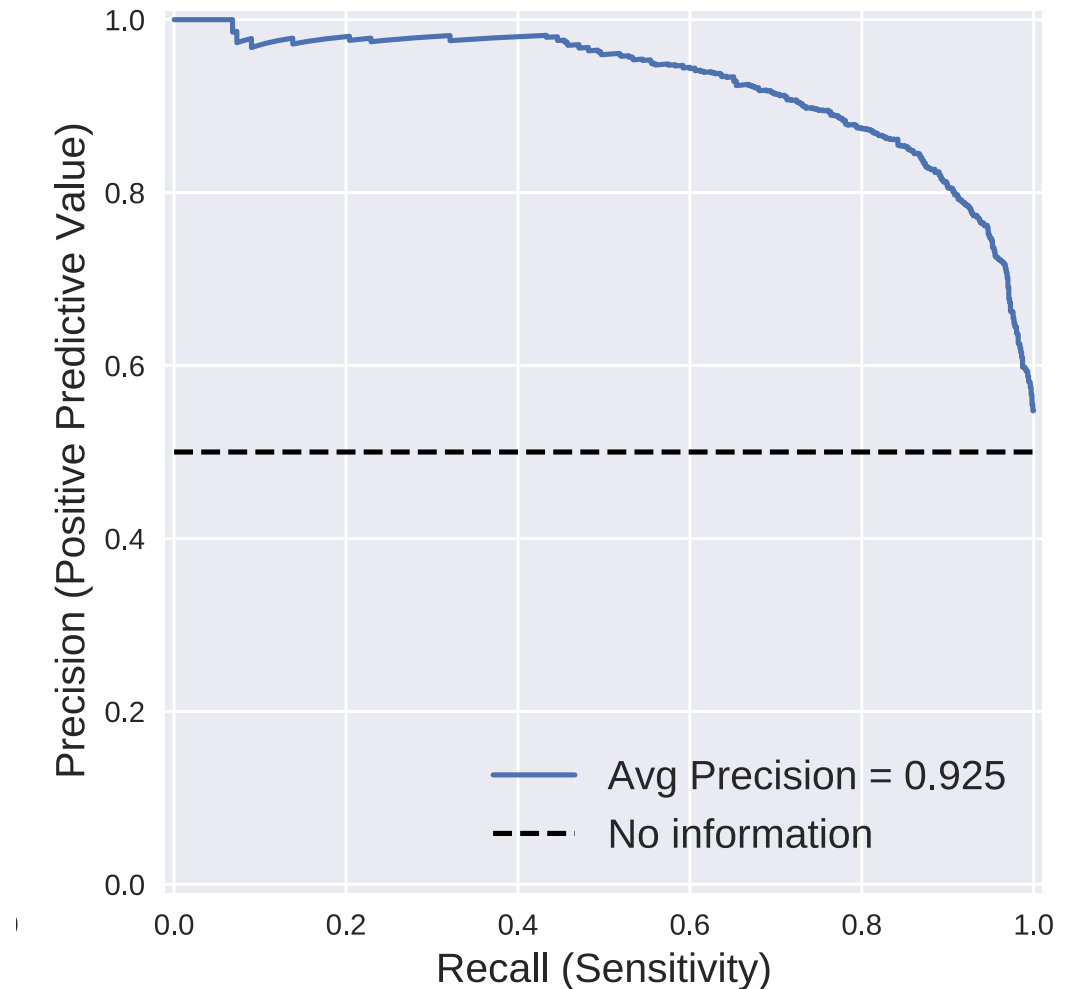
1. A computer vision model that detects carcinoma

Operating Point:
high sensitivity

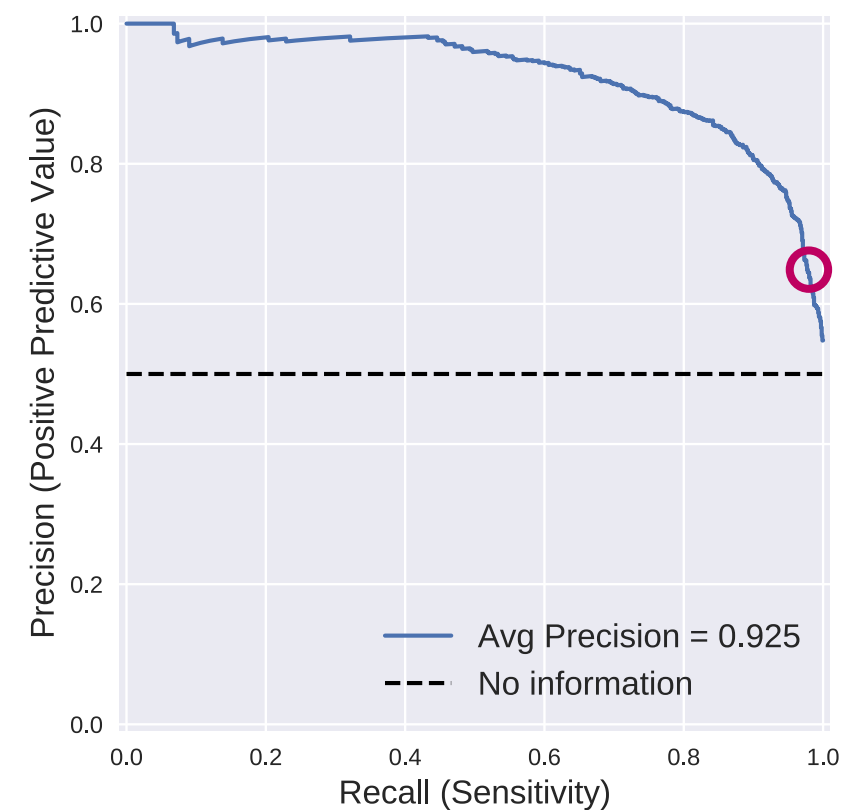
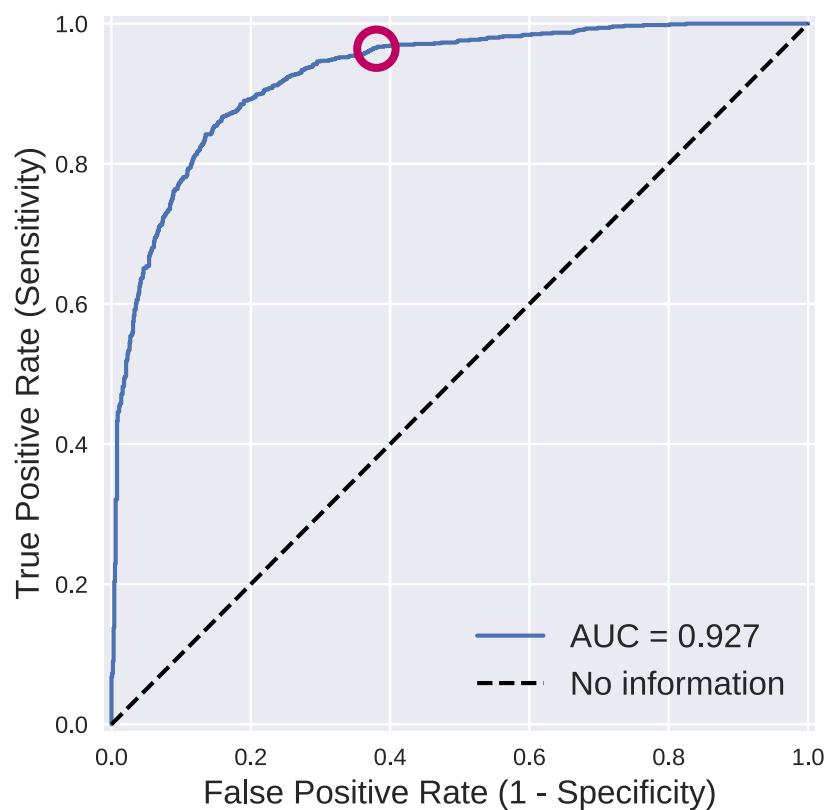
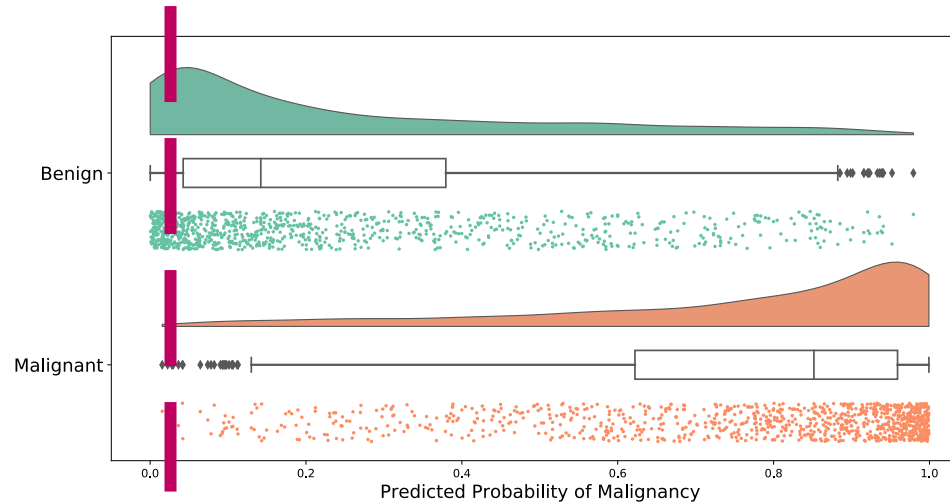


The Precision-Recall Curve (i.e., PPV-Sensitivity Curve)

- Some argue it has greater clinical relevance than the ROC curve
- The *no information* classifier always achieves PPV equal to the *base rate*, or *prevalence* (why?)
- PPV as well as the area under this curve (average precision) must be interpreted relative to prevalence



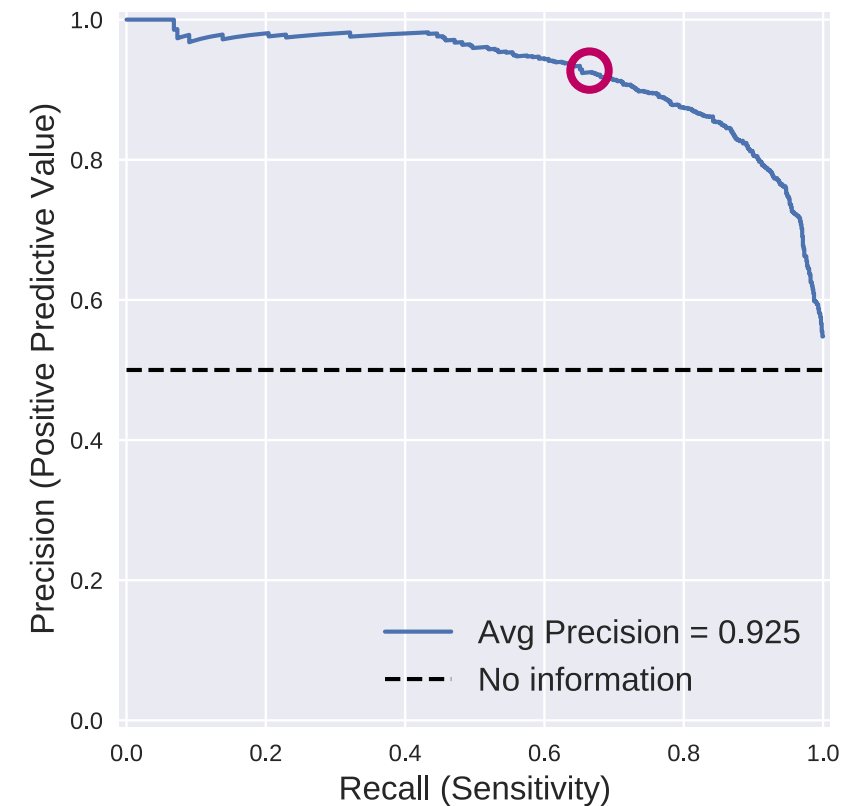
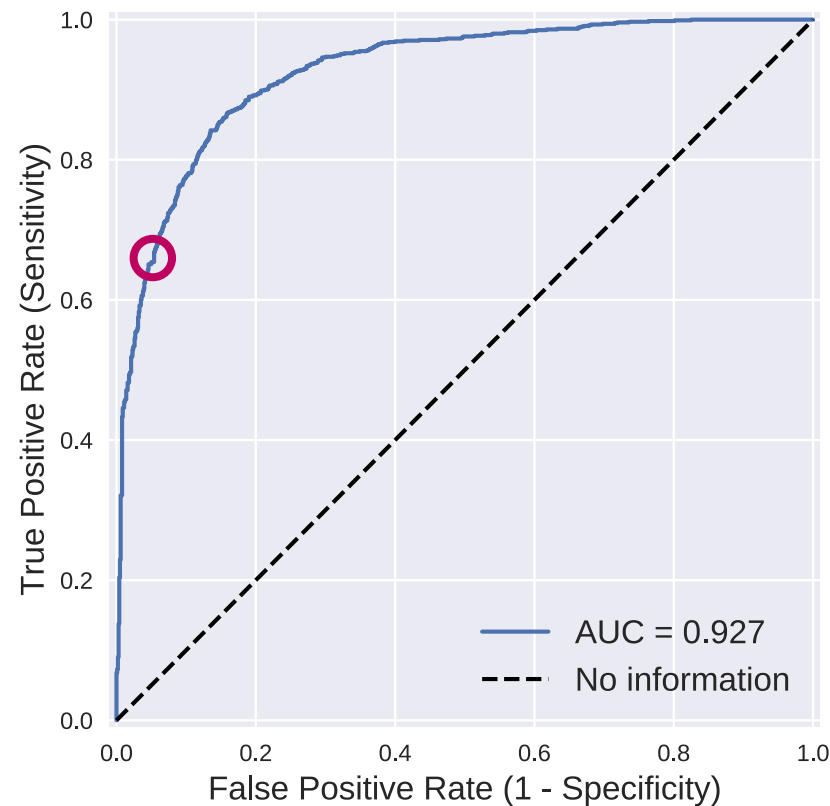
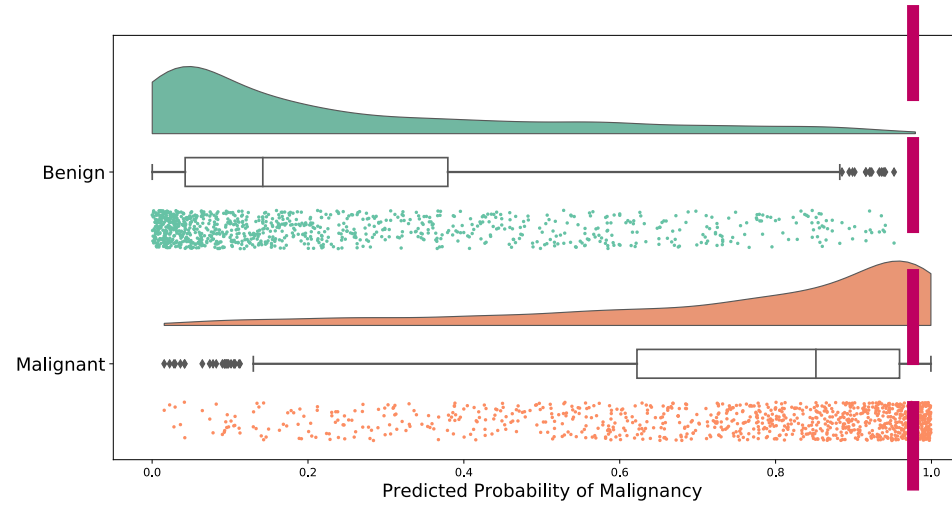
Operating Point:
high sensitivity



Healthcare Scenarios

1. A computer vision model that detects carcinoma
2. An algorithm that detects atrial fibrillation in Apple Watch users

Operating Point: *high specificity*

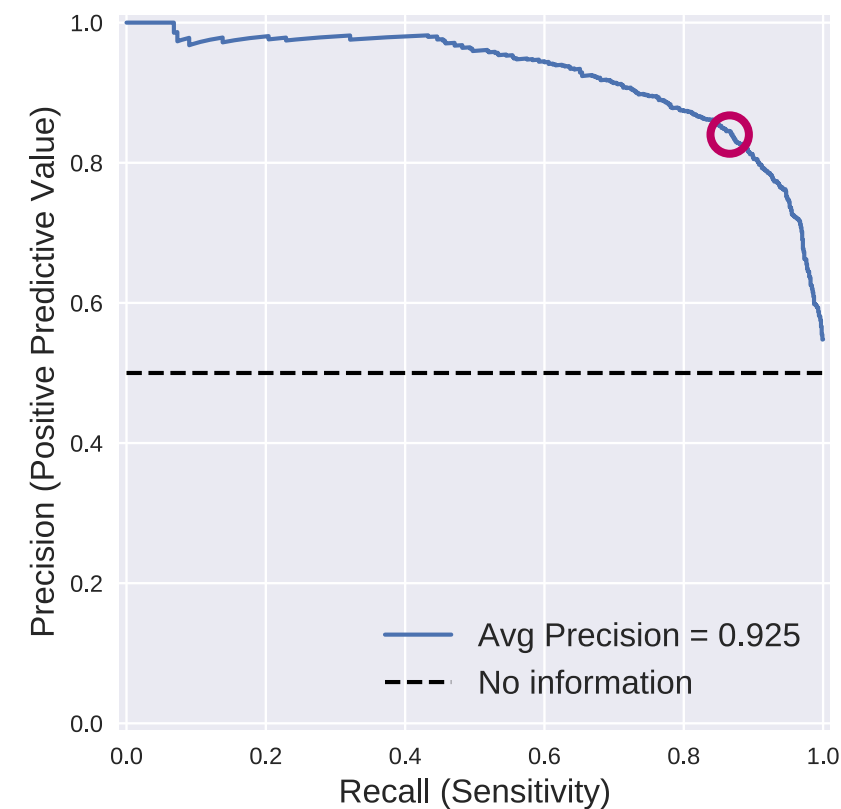
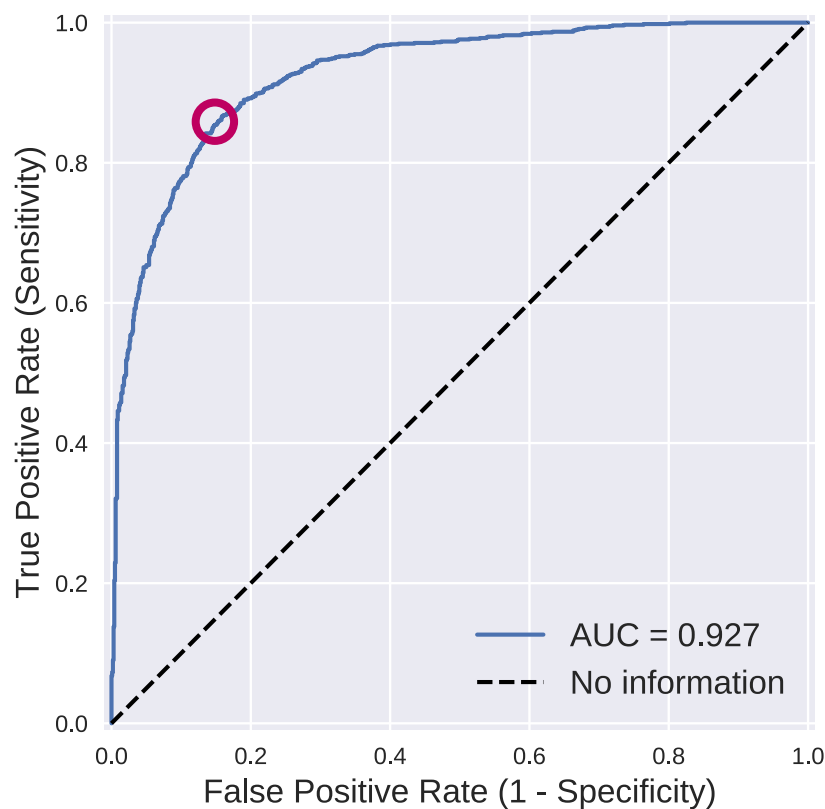
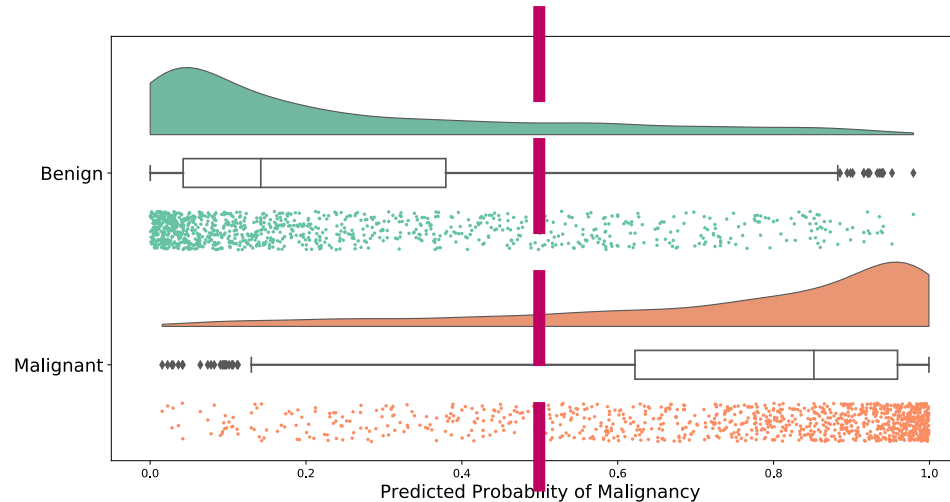


Healthcare Scenarios

1. A computer vision model that detects carcinoma
2. An algorithm that detects atrial fibrillation in Apple Watch users
3. An EHR-based model that monitors autism risk

Operating Point:

balanced



Healthcare Scenarios

1. A computer vision model that detects carcinoma
 2. An algorithm that detects atrial fibrillation in Apple Watch users
 3. An EHR-based model that monitors autism risk
- Sometimes specificity and sensitivity are difficult to interpret, particularly for rare conditions or events.
 - The most clinically relevant measure is often the positive predictive value (or negative predictive value).

Probabilistic definition of AUC

$$\text{AUC} = P(S_+ > S_-)$$

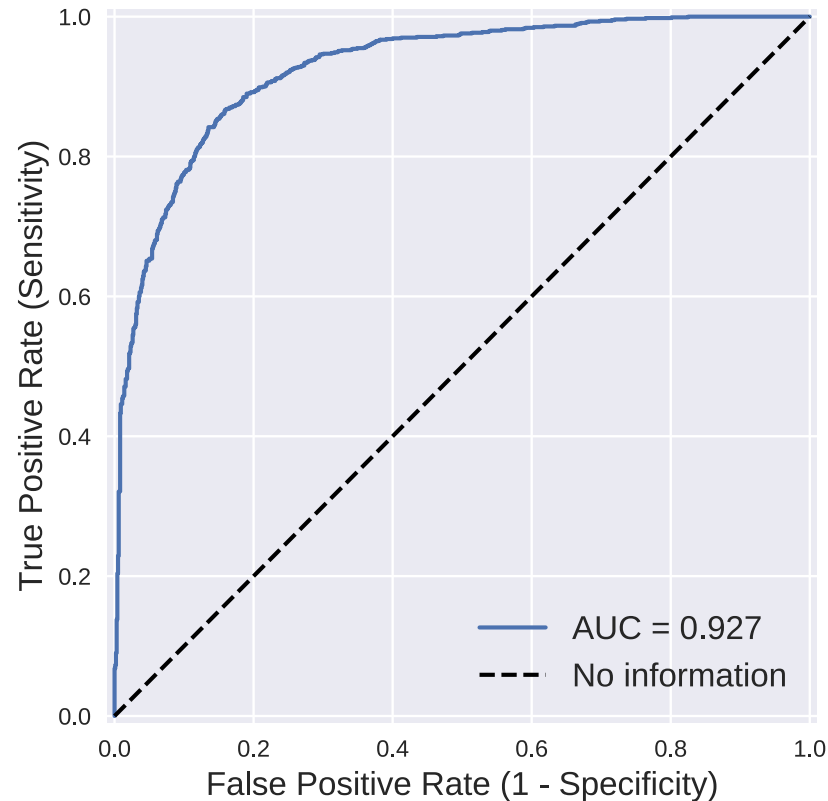
where S_+ and S_- denote random variables associated with model predictions for positive and negative outcomes, respectively.

Empirical estimator:

- Consider all pairs of samples where one is positive and one is negative
- Determine whether predictions are concordant with labels/outcomes
- Calculate % concordant

How might you code this efficiently?

Equivalence of probabilistic and graphical definitions



Consider $\int \text{TPR}(\delta) d[\text{FPR}(\delta)]$

More detailed, step by step:

Suppose our model-predicted probabilities are distributed as $p_+(t) = \Pr(T = t \mid Y = 1)$ and $p_-(t) = \Pr(T = t \mid Y = 0)$ for positive and negative samples, respectively, where $t \in (0, 1)$. Denote the corresponding CDFs as $P_+(t) = \Pr(T \leq t \mid Y = 1)$ and $P_-(t) = \Pr(T \leq t \mid Y = 0)$.

Note that the true positive rate $\text{TPR}(t) = \Pr(T > t \mid Y = 1) = 1 - P_+(t)$ and the false positive rate $\text{FPR}(t) = \Pr(T > t \mid Y = 0) = 1 - P_-(t)$.

The AUC is defined probabilistically as:

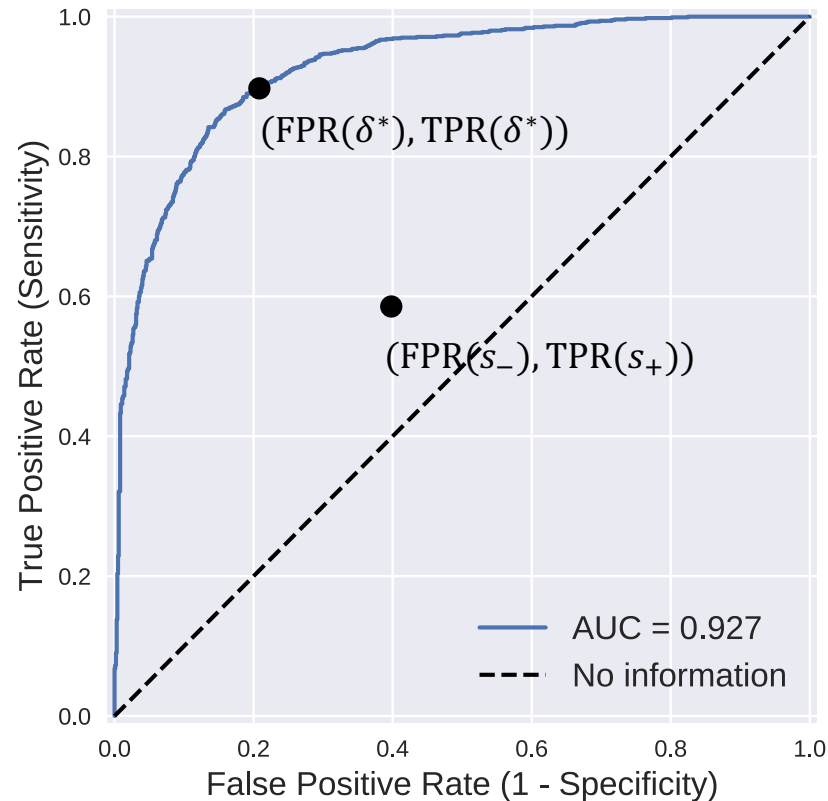
$$\begin{aligned} & \Pr(T_i < T_j \mid Y_i = 0, Y_j = 1) \\ &= \int_0^1 \Pr(T_j > t \mid Y_j = 1) \Pr(T_i = t \mid Y_i = 0) dt \\ &= \int_0^1 (1 - P_+(t)) p_-(t) dt \end{aligned} \tag{1}$$

On the other hand, the ROC curve is defined by varying the false positive rate $\tau = \text{FPR}(t)$ from 0 to 1 and determining the corresponding true positive rate $\text{TPR}(\text{FPR}^{-1}(\tau))$. The area under this curve is then given as follows:

$$\begin{aligned} & \int_0^1 \text{TPR}(\text{FPR}^{-1}(\tau)) d\tau \\ &= \int_1^0 \text{TPR}(t) \text{FPR}'(t) dt \\ &= \int_0^1 (1 - P_+(t)) p_-(t) dt, \end{aligned} \tag{2}$$

where the integration boundaries are determined by noting that $\text{FPR}(t) = 1$ for threshold $t = 0$, and $\text{FPR}(t) = 0$ for threshold $t = 1$. This establishes the correspondence between the probabilistic definition of AUC and the area under the ROC curve.

Equivalence of probabilistic and graphical definitions

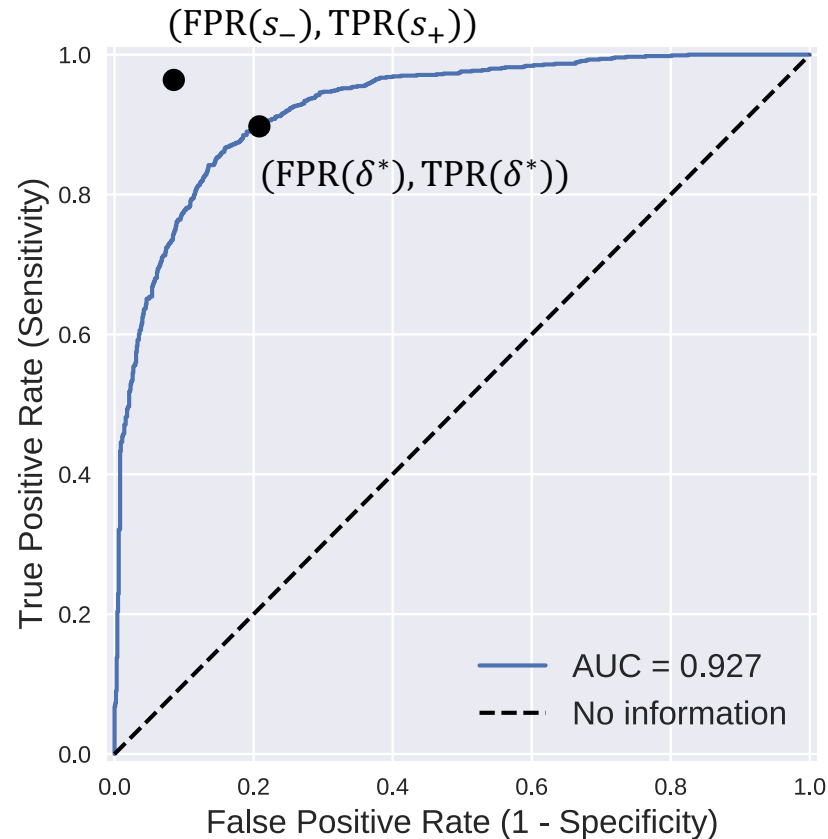


Intuition:

1. Sample predictions s_+ and s_- for positive and negative outcomes, respectively
2. Note that $\text{TPR}(S_+) \sim \text{Unif}(0,1)$ and $\text{FPR}(S_-) \sim \text{Unif}(0,1)$ (because they are 1-CDF)
3. If $s_- < s_+$, then there is $\delta^* \in (s_-, s_+)$
4. Since FPR and TPR are monotonically decreasing, $\text{FPR}(s_-) > \text{FPR}(\delta^*)$ and $\text{TPR}(s_+) < \text{TPR}(\delta^*)$

Consider $\int \text{TPR}(\delta) d[\text{FPR}(\delta)]$

Equivalence of probabilistic and graphical definitions



Consider $\int \text{TPR}(\delta) d[\text{FPR}(\delta)]$

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Report back: AUC in subgroups

Suppose we calculate AUC_A and AUC_B in subgroups A and B, respectively.

Is it necessarily true that $AUC_{A \cup B} \in (AUC_A, AUC_B)$?

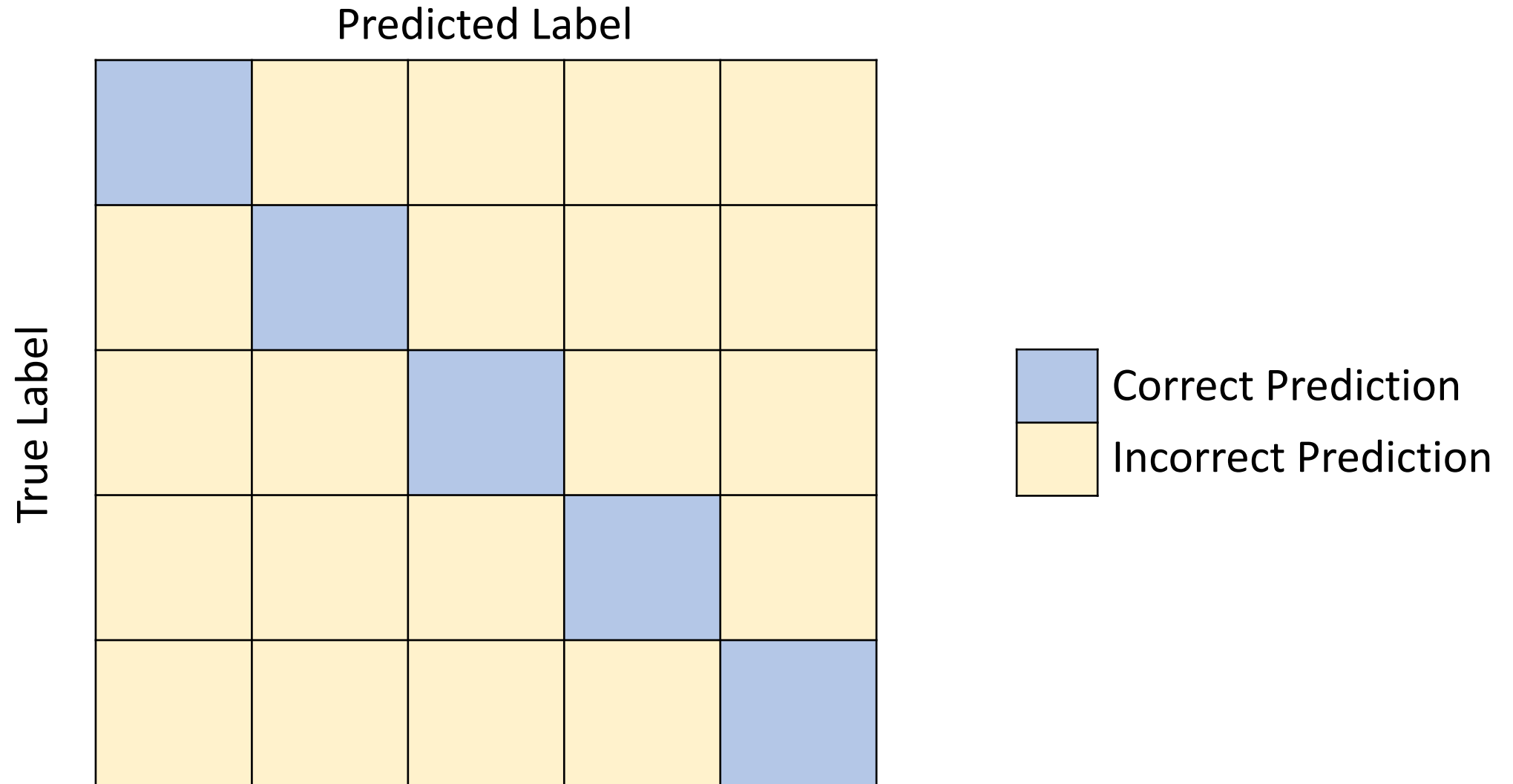
Why or why not?

Other notable properties

- AUC is rank-based and therefore invariant to monotonic transformation of predicted probabilities
 - Practical importance: AUC tells you nothing about whether predicted probabilities are accurate, i.e. calibration
- AUC is the normalized Mann-Whitney U-statistic commonly used to test whether two sets of values are drawn from the same distribution
$$\text{AUC} = U / (n_+ n_-)$$
- AUC does not depend on class balance (why?)

Multi-class problems

Multi-class problems: “Confusion Matrix”



Multi-class problems: Binary for Label 1

		Predicted Label			
True Label	True	TruePos	<--- False Negatives --->		
	False				False Pos
	True				
	False				
	True				

True Positives <--->

True Negatives

Correct Prediction

Incorrect Prediction

Multi-class problems: Binary for Label 2

		Predicted Label				
True Label						
		↑	↑	True Negatives		
		←	TruePos	False Negatives →		
		True Negatives	False Positives			
		←	←	True Negatives		

Correct Prediction

Incorrect Prediction

AUC and fairness

What does it mean for a model to be *fair*?

Consider whether predictions are fair between two groups A and B.

One possible solution is *fairness through unawareness*, also called *blindness*, wherein the model is not given information about group status.

Report back:

- What could go wrong if the model is *aware* of group status?
- What could go wrong if the model is *unaware* of group status?

Then, propose an alternative approach to achieve or evaluate fairness.

A troubled calculus

Researchers use race to build disease risk assessment tools. Can removing it help resolve medicine's race crisis?

JYOTI MADHUSOODANAN [Authors Info & Affiliations](#)

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PERSPECTIVE

Health Economics WILEY

Patient-centered appraisal of race-free clinical risk assessment

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Abstract

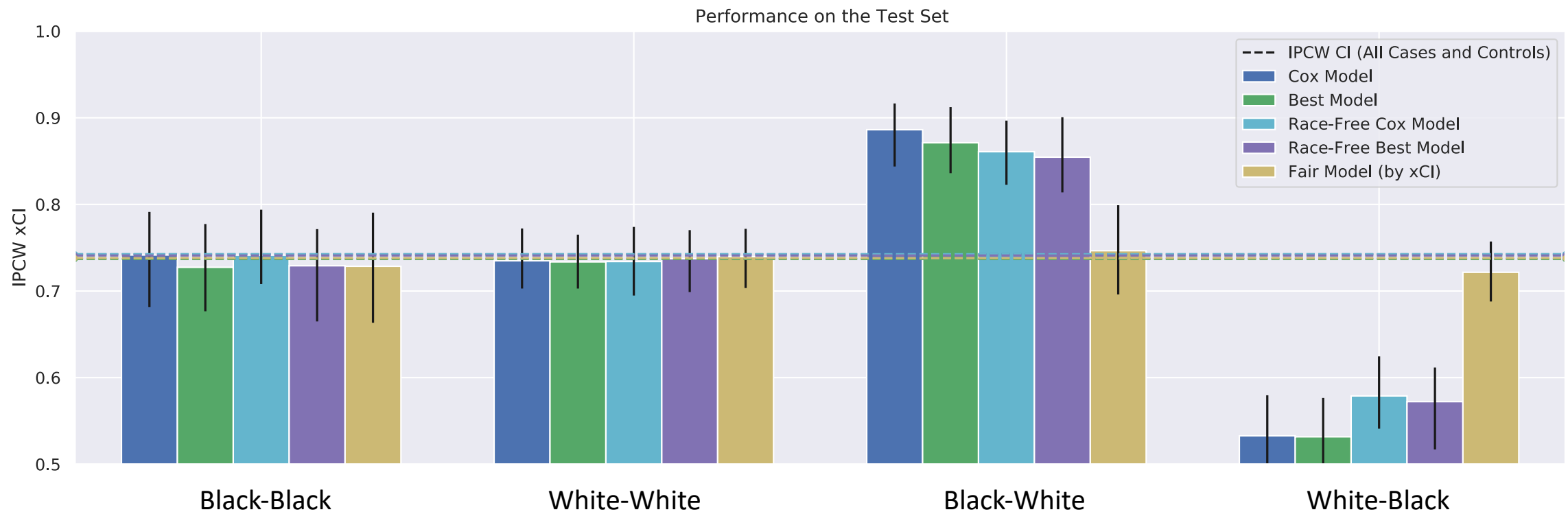
Until recently, there has been a consensus that clinicians seeking to assess patient risks of illness should condition risk assessments on all observed patient covariates with predictive power. The broad idea is that knowing more about patients enables more accurate predictions of their health risks and, hence, better clinical decisions. This consensus has recently unraveled with respect to a specific covariate, namely race. There have been increasing calls for race-free risk assessment, arguing that using race to predict health risks contributes to racial disparities and inequities in health care. In some medical fields, leading institutions have recommended race-free risk assessment. An important open question is how race-free risk assessment would affect the quality of clinical decisions. Considering the matter from the patient-centered perspective of medical economics yields a disturbing conclusion: Race-free risk assessment would harm patients of all races.

KEYWORDS

predicting illness risk, racial health disparities, risk assessment algorithms

Blindness can be ineffective or even harmful.

- In some cases it is effective.
- In other cases it isn't enough to fix the problem.
- In still other cases, it can make the problem worse.



Possible definitions of algorithmic fairness

Consider whether predictions are fair between two groups A and B.

1. Blindness: The model *cannot see* group status
2. Independence: Model predictions are *independent* of group status
3. Separation: Model *performance* is similar in all groups
4. Individual fairness: The model makes *similar predictions* for similar individuals regardless of group status

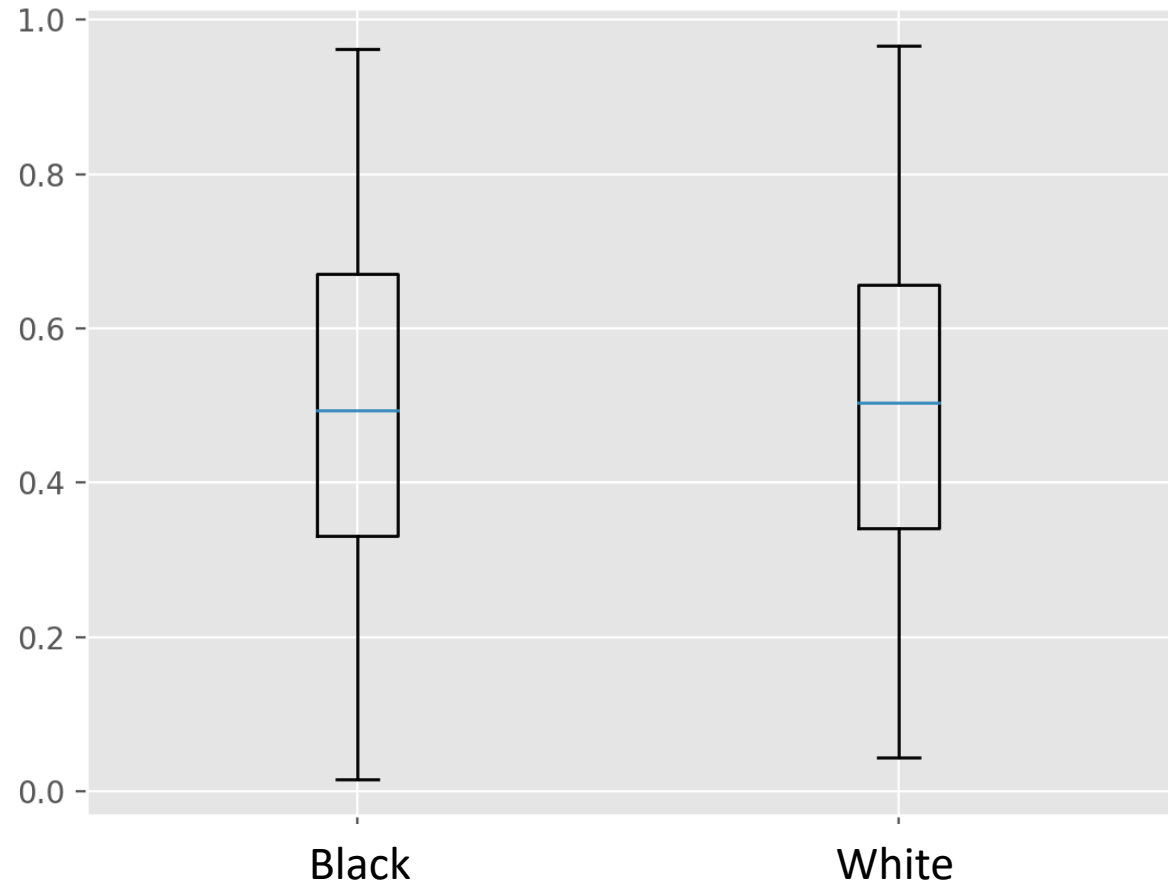
Except under special (& not realistic) circumstances, (1) – (4) are mutually exclusive.

Zooming in on *Independence*

Model predictions are independent of group status.

This implies that the distribution of model predictions is the same in each group.

Is this **fair**?

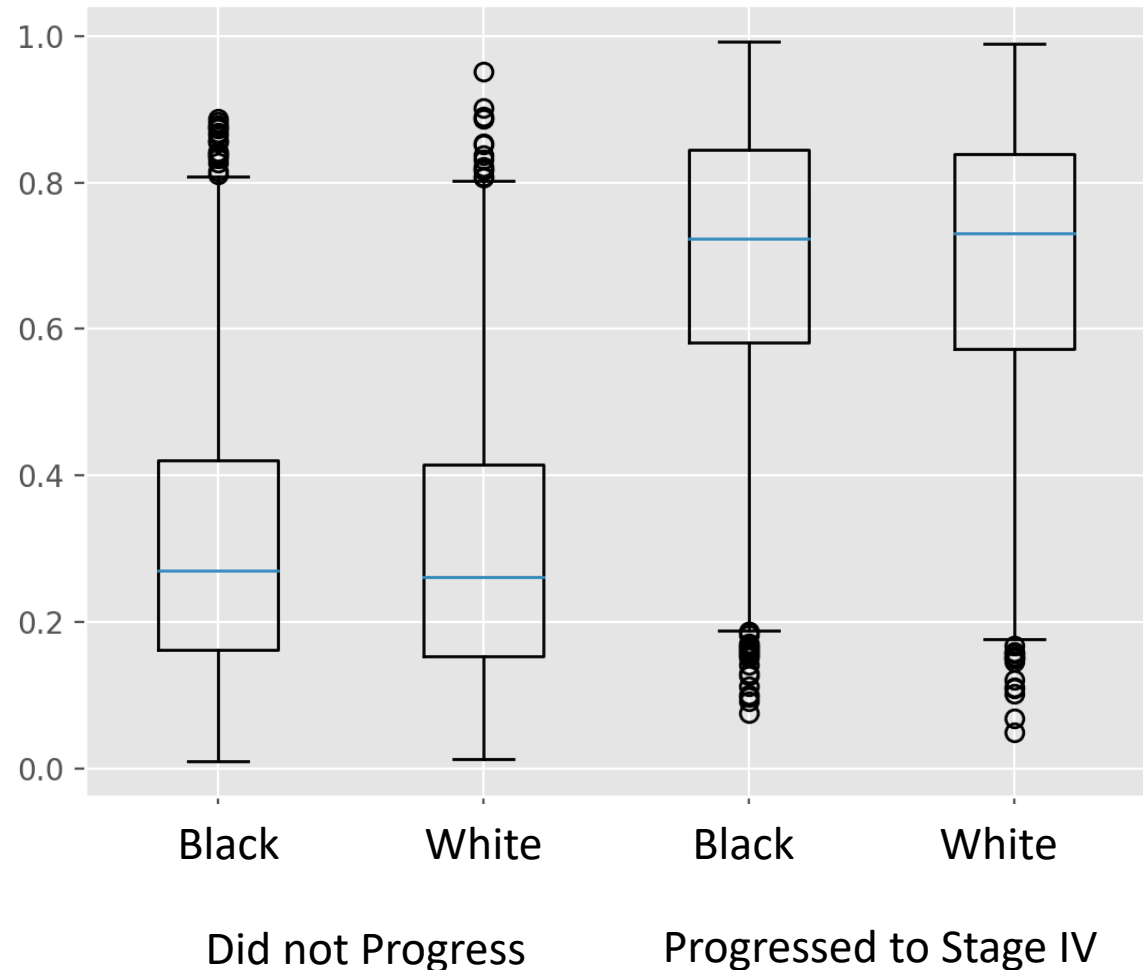


Zooming in on *Separation*

Model predictions are independent of group status *after conditioning on the outcome*.

This implies when considering individuals *with the same outcome*, the distribution of model predictions is the same in each group.

Is this **fair**?



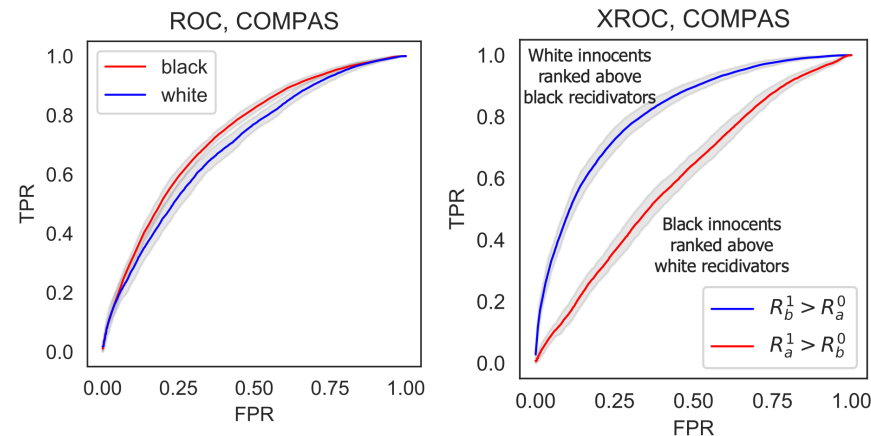
Assessing fairness with group conditional AUC (xAUC)

$$\text{xAUC}_{A,B} = P(S_+^A > S_-^B)$$

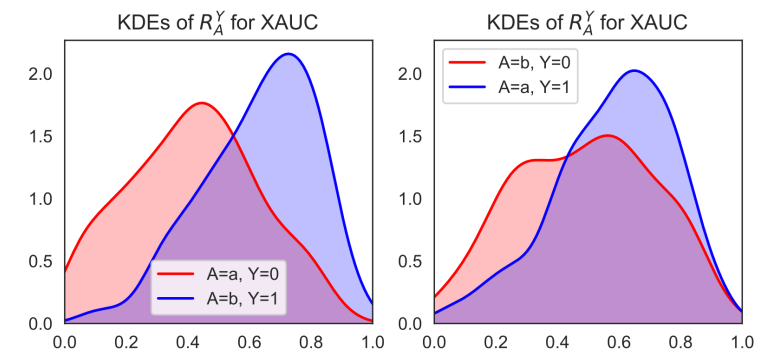
[i.e. we are now conditioning on group status]

Consider $\Delta\text{xAUC} = |\text{xAUC}_{B,A} - \text{xAUC}_{A,B}|$

Does lower ΔxAUC
imply a fairer model?



(a) ROC curves and XROC curves for COMPAS



(b) Score distributions $\Pr[R = r \mid A = a, Y = y]$